

Quantitative Risk Management

MATH 510

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Definitions

Theorems

This course is based on the ETH course and textbook of the same name, by McNeil, Fray, and Embrechts. The present document is based on this textbook and Johanna Neshlehov's lectures at McGill.

We would like to quantify one's exposure to bad financial consequences. The likelihood of loss or less-than-expected gains is called risk, and we will primarily focus on two measures of risk: value-at-risk (VaR) and expected shortfall (ES).

As a thru-line, we will study time series models, extreme value theory, and backtesting with the aim of developing accurate and robust risk measure forecasts. We will follow this by a short discussion of multivariate models.

The following are some types of risk:

Risk	Description
Credit	Odds a debtor defaults on payment
Market	Exposure to price fluctuations of bonds, stocks, or derivatives
Operational	Risk relating to circumstantial adverse events (e.g. institutional fraud)
Liquidity	Risk of damage from not having sufficient assets to pay off debts
Model	Risk associated with financial model inaccuracies; closely related to operational risk
Underwriting	Odds that an insured makes a claim on their policy

Risk is necessary:
taking this course
incurs a risk of a
poor grade

I Risk Measures

We will assume good working knowledge of probability and statistics. In addition, we remind ourselves of the *survivor function* $\bar{F}(x) := \mathbb{P}(X > x) = 1 - F(x)$, as well as the *α -quantile*, where $\alpha \in (0, 1)$, defined to be

DEF 1.1

DEF 1.2

$$q_\alpha = F^{-1}(\alpha) = \inf\{x \in \mathbb{R} : F(x) \geq \alpha\}$$

RISK FACTORS

Portfolios may include stocks, bonds, derivatives, risky loans, or insurance contracts, for example.

A *portfolio* is a collection of assets or liabilities. Denote by V_t the *value* of the portfolio at time t . We denote by Δt a *time horizon*, i.e. a duration of time. Assuming that V_t is known, that the composition of the portfolio remains constant over the time horizon, and that there are no payments made, we denote by $V_{t+\Delta t}$ the value of the portfolio at time $t + \Delta t$. Note that $V_{t+\Delta t}$ is a random variable.

DEF 1.3

DEF 1.5

Hence, $V_{t+k\Delta t}$ is the value of the portfolio at time $t + k\Delta t$

We write $\Delta V_{t+1} = V_{t+1} - V_t$. This is a random variable which takes a negative value on losses and a positive value on profits. We say that it follows the profit-and-loss distribution. Since we care about managing risk in this course, we prefer that losses are positively indicated. Define *loss* as

DEF 1.6

$$L_{t+1} = -\Delta V_{t+1} = \begin{cases} V_t - V_{t+1} & \Delta t \text{ is a short horizon} \\ V_t - \frac{V_{t+1}}{1+r_t} & \Delta t \text{ is a long horizon} \end{cases}$$

From now on, when modeling loss, we assume that V_t is realized.

Note that, for long time horizons, we must account for the time value of money. If r_t is the risk-free interest rate, applied over Δt , then $V_t(1 + r_t)$ at time $t + 1$ is equivalent in value to V_t at time t . In t dollars, then, V_{t+1} is worth $\frac{V_{t+1}}{(1+r_{t,1})}$. Working in $t + 1$ dollars, one could equivalently write $L_{t+1} = V_t(1 + r_{t,1}) - V_{t+1}$ for long time intervals.

V_t is typically a function of multiple *risk factors*, denoted by $\mathbf{Z}_t = \langle Z_{t,1}, \dots, Z_{t,d} \rangle$. We write $V_t = f(t, \mathbf{Z}_t)$ with $f : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}$, called the *risk function*. Risk factors typically change over time. We call changes in their value *risk factor changes*, denoted by $\mathbf{X}_{t+1}$. In particular,

DEF 1.7

DEF 1.8

DEF 1.9

$$\mathbf{X}_{t+1} = \mathbf{Z}_{t+1} - \mathbf{Z}_t$$

$$L_{t+1} = f(t, \mathbf{Z}_t) - f(t+1, \mathbf{Z}_t + \mathbf{X}_{t+1})$$

PROP 1.1

By [Def 1.8](#), $L_{t+1} = f(t, \mathbf{Z}_t) - f(t+1, \mathbf{Z}_{t+1}) = f(t, \mathbf{Z}_t) - f(t+1, \mathbf{Z}_t + \mathbf{X}_{t+1})$ □

PROOF.

"Observable" =
"Known" =
"Realized" =
"Actualized"

When risk factors are observed, we call $f(t, \mathbf{z}_t)$ the *realized value* of V_t . Similarly, when risk factor changes are additionally known, L_{t+1} is the *realized loss*.

DEF 1.10

DEF 1.11

Eg. 1.1 Consider a portfolio of d stocks, and let λ_i denote the number of shares in stock i at time t . Let $S_{t,i}$ denote the price of the stock i . Suppose we let our risk factors be

$$Z_{t,i} = \ln(S_{t,i}) : i \in [d]$$

DEF 1.12

so that our risk factor *changes* are $X_{t+1,i} = \ln(S_{t+1,i}) - \ln(S_{t,i}) = \ln\left(\frac{S_{t+1,i}}{S_{t,i}}\right)$. In this use case, we call X_{t+1} *log returns*. The value of the portfolio is then

$$V_t = \sum_{i=1}^d \lambda_i S_{t,i} = \sum_{i=1}^d \lambda_i e^{Z_{t,i}}$$

So that our loss, assuming that Z_t is realized, is

$$\begin{aligned} L_{t+1} &= \sum_{i=1}^d \lambda_i e^{z_{t,i}} - \sum_{i=1}^d \lambda_i e^{Z_{t+1,i}} = \sum_{i=1}^d \lambda_i e^{z_{t,i}} - \sum_{i=1}^d \lambda_i e^{X_{t+1,i} + z_{t,i}} \\ &= - \sum_{i=1}^d \lambda_i e^{z_{t,i}} (e^{X_{t+1,i}} - 1) = - \sum_{i=1}^d \rho_{t,i} (e^{X_{t+1,i}} - 1) = -V_t \sum_{i=1}^d w_{t,i} (e^{X_{t+1,i}} - 1) \end{aligned}$$

where $\rho_{t,i}$ is the total value and $w_{t,i} = \frac{\lambda_i S_{t,i}}{V_t}$ the relative weight of stock i at time t . We can write the risk function as

$$f(t, \mathbf{Z}_t) = \sum_{i=1}^d e^{Z_{t,i}}$$

DEF 1.13

DEF 1.14

Given the realized value at time t , as a function of observed risk factor changes $\mathbf{x}_{t+1} =: \mathbf{x}$, we notate $l_{[t]}(\mathbf{x}) = L_{t+1}$, which we call the *loss operator*. We say that $l_{[t]}(\mathbf{x})$ (or L_{t+1} , when it is understood that $\mathbf{z}_t$ is known) follows the *loss distribution*, which is deterministically dependent on the underlying distribution of $\mathbf{x}$.

We distinguish between two types of distributions for the risk factor changes (and, by extension, two types of loss distributions).

DEF 1.15

1. The conditional distribution of risk factor changes X_{t+1} , as a function of all information up to time t , which we denote by the sigma field $\mathcal{F}_t$. We call the resulting loss distribution the *conditional loss distribution*.

Note that an estimation of risk factor changes which simply relies on historical data is not necessarily a conditional distribution. We must be computing actual, conditional probabilities for this definition to apply.

DEF 1.16

2. The unconditional distribution of risk factor changes. The resulting loss distribution is called the *stationary loss distribution*.

The loss distribution is typically right-skewed and has a heavy tail: dramatic losses not unlikely, and heavy profits are less common than heavy losses. Thus, the normal distribution is not always a good proxy for risk factor changes (but, instead, say, a skewed student-t).

Stylized Loan Portfolio

This is an example of a similar flavor to [Example 1.1](#).

We will consider a *stylized loan portfolio*, which is a collection of loans, with interest, we make out to obligors. Let the time horizon Δt be one year (with this horizon, we should account for the time value of money).

DEF 1.17

We lend to m obligors. For each obligor i , let k_i denote the amount to be paid at the end of the present time period, consisting of the loan principle plus interest. Since this payment is made at time $t + 1$, $e_i = \frac{k_i}{1+r_t}$, called the "exposure," is the present-day money value of this payment, where r_t is the risk-free interest rate.

There is a possibility that an obligor defaults. Let $Y_{t,i}$ be an indicator variable, called the "default state," that detects whether an obligor defaults by time t . We will assume $Y_{t,i} = 0$ and $\mathbb{E}[Y_{t+1,i}] = p_i$. In case of default, the lender may recover a portion of the loan, i.e. $(1 - \delta_i)k_i$, where $\delta_i \in (0, 1]$. The expected shortfall associated with obligor i is the difference between [the present-dollar value of the loan payment at time $t + 1$] and [the present-dollar expected loan payment at time $t + 1$]

$$\frac{k_i}{1+r_t} - \underbrace{\left[p_i(1 - \delta_i) \frac{k_i}{1+r_t} + (1 - p_i) \frac{k_i}{1+r_t} \right]}_{e_i(1-p_i\delta_i)} = p_i\delta_i \frac{k_i}{1+r_t} = p_i\delta_i e_i$$

The value of the loan itself, then, is the [present-dollar loan payment's value] minus the [expected shortfall], or $e_i - p_i\delta_i e_i$. Note that this is exactly [the present-dollar expected loan payment at time $t + 1$], which we already computed. The value of whole the portfolio is thus given by

$$V_t = \sum_{i=1}^m e_i - p_i\delta_i e_i = \sum_{i=1}^m e_i(1 - p_i\delta_i)$$

At time $t + 1$, we assume that all obligors have paid off the principle. Hence, V_{t+1} encapsulates the actualized payoff made at time $t + 1$:

$$V_{t+1} = \sum_{i=1}^m Y_{t,i}(1 - \delta_i)k_i + (1 - Y_{t,i})k_i = k_i(1 - Y_{t,i}\delta_i)$$

Making a final adjustment for the time value of money, the loss of a stylized loan portfolio is

$$L_{t+1} = V_t - \frac{V_{t+1}}{1+r_t} = \sum_{i=1}^m e_i(1 - \delta_i p_i) - e_i(1 - Y_{t,i}\delta_i) = \sum_{i=1}^m e_i\delta_i(Y_{t,i} - p_i)$$

SIMPLE MODELS OF LOSS

We aim to estimate the loss distribution L_{t+1} , especially when f is unknown.

Linearized Loss

By applying the Taylor series, we obtain *linearized loss*, denoted L_{t+1}^Δ , and given by

DEF 1.18

$$L_{t+1}^\Delta = -f_t(t, \mathbf{z}_t) - \sum_{i=1}^d f_{z_{t,i}}(t, \mathbf{z}_t) X_{t+1,i}$$

However, if the loan payments continued into a second year, we would *add on* to V_{t+1} an estimation of the expected loan payment at time $t + 2$.

PROP 1.2 Assuming differentiability of f along t and $\mathbf{Z}_t$, the above definition makes sense.

PROOF.

Recall $L_{t+1} = f(t, \mathbf{z}_t) - f(t+1, \mathbf{z}_{t+1})$. We approximate $f(t+1, \mathbf{z}_{t+1})$ at the point $\langle t, \mathbf{z}_t \rangle$:

$$f(t+1, \mathbf{z}_{t+1}) \approx f(t, \mathbf{z}_t) + \nabla f(t, \mathbf{z}_t) \cdot (\langle t+1, \mathbf{z}_{t+1} \rangle - \langle t, \mathbf{z}_t \rangle)$$

This is

$$f(t, \mathbf{z}_t) + f_t(t, \mathbf{z}_t) \cdot 1 + \sum_{i=1}^d f_{z_{t,i}}(t, \mathbf{z}_t) X_{t+1,i}$$

Applying to L_{t+1} gives

$$L_{t+1}^\Delta = f(t, \mathbf{z}_t) - \left[f(t, \mathbf{z}_t) + f_t(t, \mathbf{z}_t) + \sum_{i=1}^d f_{z_{t,i}}(t, \mathbf{z}_t) X_{t+1,i} \right] = -f_t(t, \mathbf{z}_t) - \sum_{i=1}^d f_{z_{t,i}}(t, \mathbf{z}_t) X_{t+1,i}$$

as desired. $\square$

Linearized loss is convenient and easy to understand, but it relies on heavy differentiability assumptions and is accurate only locally.

Fig. 1.2 Following [Example 1.1](#), we use $f(t, \mathbf{z}_t)$ to compute the linearized loss:

$$\begin{aligned} L_{t+1}^\Delta &= -0 - \sum_{i=1}^d \lambda_i e^{Z_{t,i}} X_{t+1,i} = - \sum_{i=1}^d \lambda_i e^{Z_{t,i}} X_{t+1,i} \\ &= -V_t(\mathbf{w}_t \cdot \mathbf{X}_{t+1}) = -\boldsymbol{\rho}_t \cdot \mathbf{X}_{t+1} \end{aligned}$$

PROP 1.3 If we assume that $\mathbf{X}_{t+1} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ follows the multivariate normal distribution, then, under linearized loss, $L_{t+1}^\Delta \sim \mathcal{N}(-\boldsymbol{\rho}_t \cdot \boldsymbol{\mu}, \boldsymbol{\rho}_t^T \Sigma \boldsymbol{\rho}_t) = \mathcal{N}(-V_t \mathbf{w}_t \cdot \boldsymbol{\mu}, V_t^2 \mathbf{w}_t^T \Sigma \mathbf{w}_t)$.

PROOF.

Under the $L_{t+1}^\Delta = -\boldsymbol{\rho}_t \cdot \mathbf{X}_{t+1}$ view, L_{t+1}^Δ is a sum of normals, and hence normal. We compute $\mathbb{E}[L_{t+1}] = -\mathbb{E}[\sum_{i=1}^d \rho_{t,i} X_{t+1,i}] = -\boldsymbol{\rho}_t \cdot \boldsymbol{\mu} = -V_t \mathbf{w}_t \cdot \boldsymbol{\mu}$. We compute variance similarly. $\square$

In the following sections, we will consider methods for estimating the loss distribution of a stock portfolio (but applicable generally), per [Example 1.1](#).

Variance-Covariance Method

Utilizing the discussion above, assume that $\mathbf{X}_{t+1} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, where we compute $\boldsymbol{\mu}$ and Σ via any suitable estimator (e.g. MLE), based on historical data $\mathbf{x}_{t-n+1}, \dots, \mathbf{x}_t$. Then, with $L_{t+1}^\Delta \sim \mathcal{N}(-\boldsymbol{\rho}_t \cdot \boldsymbol{\mu}, \boldsymbol{\rho}_t^T \Sigma \boldsymbol{\rho}_t)$, we call the approximation $L_{t+1} \approx L_{t+1}^\Delta$ the *variance-covariance method*. This method is exclusive to a stock portfolio setting, and generates a stationary distribution.

DEF 1.19

Historical Simulation Method

Given historical data $\mathbf{x}_{t-n+1}, \dots, \mathbf{x}_t$, we can simulate possible future losses, assuming that the risk function f is available to us:

$$\tilde{L}_s = f(t, \mathbf{z}_t) - f(t+1, \mathbf{z}_t + \mathbf{x}_s) : s \in [t-n+1, t]$$

Then, we average these estimates to yield an approximation of the loss distribution

$$L_{t+1} \approx L_{t+1}^* : \mathbb{P}(L_{t+1}^* \leq x) \approx \frac{1}{n} \sum_{s=1}^n \mathbb{1}(\tilde{L}_{t-n+s} \leq x)$$

which we call the *historical simulation method*.

DEF 1.20

If $\mathbf{X}_s$ are IID, the convergence of the historical simulation method to the true loss distribution is ensured, but this is very rarely the case in real life. As with the variance-covariance method, this generates a stationary distribution.

Sometimes, when f is not available to us, we may simply use observed losses, but this is a crude method: we are simply averaging past losses to determine future performance.

Monte Carlo Method

We create a model for risk factor changes, which we "tune" using historical data $\mathbf{x}_{t-n+1}, \dots, \mathbf{x}_t$. Then, using this model, we construct an arbitrary number M of new data points, $\mathbf{x}_{t+1}^{(1)}, \dots, \mathbf{x}_{t+1}^{(M)}$. Using this data, we simulate future losses, similar to the historical simulation method

$$\tilde{L}_{t+1}^{(m)} = f(t, \mathbf{z}_t) - f(t+1, \mathbf{z}_t + \mathbf{x}_{t+1}^{(m)}) : m \in [M]$$

We average these estimates to yield an approximation of the loss distribution

$$L_{t+1} \approx L_{t+1}^* : \mathbb{P}(L_{t+1}^* \leq x) \approx \frac{1}{M} \sum_{m=1}^M \mathbb{1}(\tilde{L}_{t+1}^{(m)} \leq x)$$

which we call the *Monte Carlo method*

DEF 1.21

By selecting a large M or providing alternative models, this method can potentially circumvent a lack of historical data. However, this method can be expensive, as it relies on training and sampling from a model. We can use the Monte Carlo method to generate either a conditional or stationary loss distribution.

When f is not available to us, we can alternatively design a model for the losses themselves, and generate $\tilde{L}_{t+1}^{(m)}$ directly. In this course, we will primarily use Monte Carlo and historical simulation methods in our analyses.

VALUE-AT-RISK AND EXPECTED SHORTFALL

Coherent Risk Measures

In order to consolidate our view of a portfolio's exposure to risk, we employ mappings from loss variables L_{t+1} to $\mathbb{R}$, called risk measures. Historically, risk measurement has been comprised of the following methods:

The historical simulation method is akin to driving by only looking through the rear view mirror

The Monte Carlo Method is akin to driving by using the rear view mirror to view a front-facing funny mirror. Slightly better.

Notational Risk is simply the sum total of the values of the portfolio's securities.

Scenario-Based One predicts (e.g. via Markov chains) future scenarios and associated probabilities. One then measures the expected maximum future loss of the portfolio against these scenarios.

Loss-Based One designs and trains a statistical, time series model of the loss distribution. One then finds statistical proxies for risk (e.g. 95th percentile loss).

Eg. 1.3 Using the notational method, we could measure, say, the average gross income over the past 3 years, excluding years where incoming is 0 or negative. This makes sense for operational risk (i.e. risk of loss by internal mismanagement), which is proportional to active cash flows. Explicitly, we have

$$\alpha \frac{\sum_{i=1}^3 \max(G_{t-i}, 0)}{\sum_{i=1}^3 \mathbb{1}(G_{t-i} > 0)}$$

where $\alpha \approx 0.15$ and G_t is gross annual income for year t . In the industry, this is referred to as the "basic indicator approach."

Eg. 1.4 Using the scenario-based method, we consider some fixed risk factor changes $\mathbf{x}_1, \dots, \mathbf{x}_n$, with associated weights $w_1, \dots, w_n$ describing their likelihood of occurring. A scenario-based risk assessment might be

$$\psi(L) = \max\{w_i l_{[t]}(\mathbf{x}_i)\}_{i \in [n]}$$

Eg. 1.5 The oldest loss-based measure simply used the variance of the $P\&L$ distribution as a risk measure. However, this approach requires the existence of a second moment. We will develop more robust methods.

DEF 1.22 A *risk measure* is a function $\rho : \mathcal{M} \rightarrow \mathbb{R} : L \mapsto \rho(L)$ on the space of random variables $\mathcal{M}$ representing losses. We frequently envisage $\rho(L)$, up to rescaling, as the total amount of equity capital required to back a position with loss function L .

DEF 1.23 A risk measure ρ is called *coherent* if it satisfies

Monotonicity For all $L_1, L_2 \in \mathcal{M}$ with $L_1 \geq L_2$ almost surely,

$$\rho(L_1) \geq \rho(L_2)$$

Recall the notion of almost surely:
 $\mathbb{P}(L_1 \geq L_2) = 1$

Invariance For any $L \in \mathcal{M}$ and $\ell \in \mathbb{R}$,

$$\rho(L + \ell) = \rho(L) + \ell$$

Subadditivity For any $L_1, L_2 \in \mathcal{M}$, we have

$$\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$$

Homogeneity For any $L \in \mathcal{M}$ and $\lambda \geq 0 \in \mathbb{R}$,

$$\rho(\lambda L) = \lambda \rho(L)$$

These axioms are informed by practice and financial intuition. Monotonicity implies that larger losses yields larger risk; invariance ensures that risk measures are constant up to a shifting of the entire loss spectrum; subadditivity captures the quality that diversified portfolios are less risky; and homogeneity ensures that variance scales with risk.

A coherent risk measure is convex, i.e. for all $L_1, L_2 \in \mathcal{M}$ and $\lambda \in (0, 1)$

PROP 1.4

$$\rho(\lambda L_1 + (1 - \lambda)L_2) \leq \lambda \rho(L_1) + (1 - \lambda)\rho(L_2)$$

However, the converse is not true.

By subadditivity and homogeneity,

PROOF.

$$\rho(\lambda L_1 + (1 - \lambda)L_2) \leq \rho(\lambda L_1) + \rho((1 - \lambda)L_2) = \lambda \rho(L_1) + (1 - \lambda)\rho(L_2)$$

□

Value-at-Risk

Given a tolerance $\alpha \in (0, 1)$, where, frequently, $\alpha \in \{0.95, 0.99\}$, *Value-at-Risk* is defined as

DEF 1.24

$$\text{VaR}_\alpha(L) = q_\alpha(F_L) = F_L^{-1}(\alpha) = \inf\{x \in \mathbb{R} : F_L(x) \geq \alpha\}$$

where F_L is the CDF of the loss distribution.

In other words, $\text{VaR}_\alpha(L)$ is the α -quantile of losses. In even more words, $\text{VaR}_\alpha(L)$ is the greatest lower bound such that seeing a loss $x \geq \text{VaR}_\alpha(L)$ will occur with probability less than $1 - \alpha$. In another set of other words, losses exist in $(-\infty, \text{VaR}_\alpha(L)]$ with probability α .

$x_0 = \text{VaR}_\alpha(L)$ if and only if $\mathbb{P}(L \leq x_0) \geq \alpha$ and $\mathbb{P}(L \leq x) < \alpha$ whenever $x < x_0$.

PROP 1.5

This is the definition of the infimum, i.e. whenever $x < x_0$, $x \notin \{x \in \mathbb{R} : F_L(x) \geq \alpha\}$, and $x_0 \in \{x \in \mathbb{R} : F_L(x) \geq \alpha\}$. □

PROOF.

$\text{VaR}_\alpha(L)$ is not subadditive, and is thus not coherent. It does satisfy the other axioms.

PROP 1.6

As proof of the fact that VaR does not satisfy subadditivity, we'll consider a worked (counter)example. It is left as an exercise to show that the other axioms are satisfied.

PROOF.

Consider a market of 50 distinct, defaultable bonds with independent defaults. The default variable for each bond is IID $Y_i \sim \text{Ber}(0.02) : i \in [50]$. The current price of bonds is 95, with a face value equal to 100. In **Portfolio A**, we buy 100 units of Bond 1. In **Portfolio B**, we buy 2 units of each bond (i.e. 2 of Bond 1, 2, ..., 50). The value of each portfolio is 9500.

Intuitively, we understand Portfolio A to be riskier, and this would be the conclusion

of a subadditive risk measure. However, we'll see that VaR suggests the opposite.

$$L_i = 95 - 100(1 - Y_i) \implies \mathbb{P}(L_i \leq x) = \begin{cases} 1 & x \leq 95 \\ 0 & x < -5 \\ 0.98 & x \in [-5, 95) \end{cases}$$

Then, with a tolerance of $\alpha = 0.95$, $\text{VaR}_\alpha(L_i) = -5$. We conclude that

$$\text{VaR}_\alpha(L^A) = \text{VaR}_\alpha(100L_1) = 100\text{VaR}_\alpha(L_1) = -500$$

Noting that $L^B = \sum_{i=1}^{50} 2L_i = -500 + 200 \sum_{i=1}^{50} Y_i$, we have that

$$L^B \sim -500 + 200\text{Bin}(50, 0.02)$$

It follows that $\text{VaR}_\alpha(L^B) = -500 + 200\text{VaR}_\alpha(\text{Bin}(50, 0.02))$, which, one can calculate, is $-500 + 200 \cdot 3 = 100$. Hence, $\text{VaR}_\alpha(L^A) < \text{VaR}_\alpha(L^B)$, which contradicts subadditivity. $\square$

Expected Shortfall

DEF 1.25 The *expected shortfall* of a loss distribution L is a risk measure defined to be

$$\text{ES}_\alpha(L) = \mathbb{E}[L | L \geq \text{VaR}_\alpha(L)]$$

In other words, assuming that we incur losses above the α -quantile, how much should we expect to lose? This is akin to a worst-case scenario estimation. For instance, if $\alpha = 0.95$, $\text{ES}_\alpha(L)$ describes the expected loss among probability < 0.05 losses.

PROP 1.7 If $L = \mu + \sigma L^*$, then $\text{VaR}_\alpha(L) = \mu + \sigma \text{VaR}_\alpha(L^*)$ and $\text{ES}_\alpha(L) = \mu + \sigma \text{ES}_\alpha(L^*)$.

PROOF.

These are axioms 2 and 4 of coherence restated. Note that expected shortfall inherits these axioms from value-at-risk (but not necessarily subadditivity). Explicitly:

$$\begin{aligned} \text{ES}_\alpha(L) &= \mathbb{E}(\mu + \sigma L^* | \mu + \sigma L^* \geq \text{VaR}_\alpha(L)) = \mathbb{E}(\mu + \sigma L^* | \mu + \sigma L^* \geq \mu + \sigma \text{VaR}_\alpha(L^*)) \\ &= \mathbb{E}(\mu + \sigma L^* | L^* \geq \text{VaR}_\alpha(L^*)) = \mu + \sigma \mathbb{E}(L^* | L^* \geq \text{VaR}_\alpha(L^*)) \\ &= \mu + \sigma \text{ES}_\alpha(L^*) \quad \square \end{aligned}$$

PROP 1.8 If L has a continuous distribution F_L , then $\text{ES}_\alpha(L)$ is a coherent risk measure.

PROOF.

We only need to treat subadditivity. Note that

$$\text{ES}_\alpha(L) = \frac{1}{1 - \alpha} \mathbb{E}(L \mathbb{1}(L \geq \text{VaR}_\alpha(L)))$$

We will show

$$(1 - \alpha) [\text{ES}_\alpha(L_1) + \text{ES}_\alpha(L_2) - \text{ES}_\alpha(L_1 + L_2)] \geq 0 \quad \star$$

Let $I_i := \mathbb{1}(L_i \geq \text{VaR}_\alpha(L_i))$ for $i = 1, 2$, and $I_{12} = \mathbb{1}(L_1 + L_2 \geq \text{VaR}_\alpha(L_1 + L_2))$. Then

$$\begin{aligned} \star &\iff \mathbb{E}(L_1 I_1 + L_2 I_2 - (L_1 + L_2) I_{12}) \geq 0 \\ &\iff \mathbb{E}(L_1(I_1 - I_{12})) + L_2(I_2 - I_{12}) \geq 0 \end{aligned}$$

We show that

$$\begin{aligned} \mathbb{E}(L_1(I_1 - I_{12})) &\geq \mathbb{E}[\text{VaR}_\alpha(L_1)(I_1 - I_{12})] = \text{VaR}_\alpha(L_1)\mathbb{E}(I_1 - I_{12}) \\ &= \mathbb{E}[I_1] - \mathbb{E}[I_{12}] = \mathbb{P}(L_1 \geq \text{VaR}_\alpha(L_1)) - \mathbb{P}(L_1 + L_2 \geq \text{VaR}_\alpha(L_1)) = 0 \end{aligned}$$

The L_2 case is symmetric. If $I_1 = 1$, then $I_1 - I_{12} \geq 0$. Conversely, if $I_1 = 0$, then $I_1 - I_{12} \leq 0$. Hence

$$\mathbb{E}((L_1 - \text{VaR}_\alpha(L_1))(I_1 - I_{12})) \geq 0 \quad \square$$

If $\mathbb{E}[|L|] < \infty$, then

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_{\text{VaR}_\alpha(L)}^{\infty} x f(x) dx$$

PROP 1.9

We can view $L|L \geq \text{VaR}_\alpha(L)$ as a variable with CDF

$$\begin{aligned} G(x) &= \mathbb{P}(L \leq x | L \geq \text{VaR}_\alpha(L)) = \begin{cases} 0 & x < \text{VaR}_\alpha(L) \\ \frac{\mathbb{P}(L \in [\text{VaR}_\alpha(L), x])}{\mathbb{P}(L \geq \text{VaR}_\alpha(L))} & x \geq \text{VaR}_\alpha(L) \end{cases} \\ &= \frac{F_L(x) - F_L(\text{VaR}_\alpha(L))}{1-\alpha} = \frac{F_L(x) - \alpha}{1-\alpha} : x \geq \text{VaR}_\alpha(L) \end{aligned}$$

PROOF.

Hence, $L|L \geq \text{VaR}_\alpha(L)$ will have PDF $\frac{f(x)}{1-\alpha}$ when $x \geq \text{VaR}_\alpha(L)$, and 0 otherwise. $\square$

Eg. 1.6 Suppose $L \sim \mathcal{N}(\mu, \sigma^2)$. We know $L \stackrel{d}{=} \mu + \sigma L^*$, where $L^* \sim \mathcal{N}(0, 1)$. Then $\text{VaR}_\alpha(L) = \mu + \sigma \text{VaR}_\alpha(L^*) = \mu + \sigma \Phi^{-1}(\alpha)$.

Eg. 1.7 Under the same assumptions, we compute $\text{ES}_\alpha(L)$. This is $\mu + \sigma \text{ES}_\alpha(L^*)$, with, by [Prop 1.9](#),

$$\text{ES}_\alpha(L^*) = \frac{1}{1-\alpha} \int_{\Phi^{-1}(\alpha)}^{\infty} \phi(x) x dx$$

where $\mathcal{N}(0, 1)$ follows the PDF ϕ . But this has the property that $\phi'(x) = -x\phi(x)$, so we conclude

$$\text{ES}_\alpha(L^*) = \frac{1}{1-\alpha} [-\phi(x)]_{\Phi^{-1}(\alpha)}^{\infty} = \frac{\phi(\Phi^{-1}(\alpha))}{1-\alpha}$$

and hence $\text{ES}_\alpha(L) = \mu + \sigma \frac{\phi(\Phi^{-1}(\alpha))}{1-\alpha}$

PROP 1.10 If F_L is continuous and $\mathbb{E}[|L|] < \infty$, then

$$\text{ES}_\alpha(L) = \int_\alpha^1 \text{VaR}_u(L) du$$

DEF 1.26 The *shortfall-to-quantile ratio* is defined to be

$$\lim_{\alpha \rightarrow 1} \frac{\text{ES}_\alpha(L)}{\text{VaR}_\alpha(L)}$$

This measures the rate at which our expected losses at a threshold grow with the threshold itself. We imagine, for heavy tailed distributions, that this is large (there is more "room" at the extremes). From a risk manager's perspective, we prefer this to be not so large.

Fig. 1.8 When $L^* \sim \mathcal{N}(0, 1)$, the shortfall-to-quantile ratio is 1. We have

$$\lim_{\alpha \rightarrow 1} \frac{\text{ES}_\alpha(L^*)}{\text{VaR}_\alpha(L^*)} = \lim_{\alpha \rightarrow 1} \frac{\frac{1}{1-\alpha} \phi(\Phi^{-1}(\alpha))}{\Phi^{-1}(\alpha)} = \lim_{x \rightarrow \infty} \frac{\frac{1}{1-\Phi(x)} \phi(x)}{x} = \lim_{x \rightarrow \infty} \frac{\frac{\phi(x)}{x}}{1 - \Phi(x)}$$

Using L'Hopital's rule,

$$\lim_{x \rightarrow \infty} \frac{\frac{\phi(x)}{x}}{1 - \Phi(x)} = \lim_{x \rightarrow \infty} \frac{1 + x^2}{x^2} = 1$$

$L \sim \mathcal{N}(\mu, \sigma^2)$, the limit will also go to 1.

Fig. 1.9 If L follows the Pareto distribution with parameter $\theta > 0$, i.e.

$$F_L(x) = 1 - x^{-\theta} : x \geq 1$$

the shortfall-to-quantile ratio is $\frac{\theta}{\theta-1}$ when $\theta > 1$, and infinite otherwise.

The PDF of L is $L(x) = \theta x^{-\theta-1} : x \geq 1$. Computing VaR_α :

$$1 - x^{-\theta} = \alpha \implies x = \text{VaR}_\alpha(L) = (1 - \alpha)^{-1/\theta}$$

Similarly,

$$\text{ES}_\alpha(L) = \frac{1}{1 - \alpha} \int_{(1-\alpha)^{-1/\theta}}^{\infty} x \theta x^{-\theta-1} dx = \frac{1}{1 - \alpha} \int_{(1-\alpha)^{-1/\theta}}^{\infty} \theta x^{-\theta} dx$$

Note that this integral only converges when $\theta > 1$. Computing the integral,

$$\text{ES}_\alpha(L) = \frac{\theta}{\theta - 1} (1 - \alpha)^{-1/\theta} = \frac{\theta}{\theta - 1} \text{VaR}_\alpha(L)$$

Hence, $\frac{\text{ES}_\alpha(L)}{\text{VaR}_\alpha(L)} = \frac{\theta}{\theta-1}$

Eg. 1.10 If $L \sim t_\nu(\mu, \sigma^2)$, i.e. a student-t distribution with mean μ and variance σ^2 , the shortfall-to-quantile ratio is $\frac{\nu}{\nu-1}$ when $\nu > 1$, and infinite otherwise. We do not prove this here.

Broadly speaking, expected shortfall provides more information about the tail distribution than value-at-risk, though, if estimating empirically, it requires more data. Expected shortfall has the added benefit of being coherent (i.e. subadditive) under continuity.

Recall that the loss distribution, as we first introduced them, is the distribution of losses *at time* $t + 1$, where we have access to information up to time t , $\mathcal{F}_t$. Thus, these measures are naturally dependent on a forecast, as a random variable, of future losses. In the next section, we will discuss time series models that aim to provide faithful loss forecasts. Ultimately, they will be used to compute VaR and ES, as described here.

II Financial Time Series

STATISTICAL BACKGROUND

Autocorrelation

DEF 2.1 A *time series* is a collection $(X_t : t \in \mathbb{Z})$ of random variables, or simply just X_t . We will frequently look at log returns as an example.

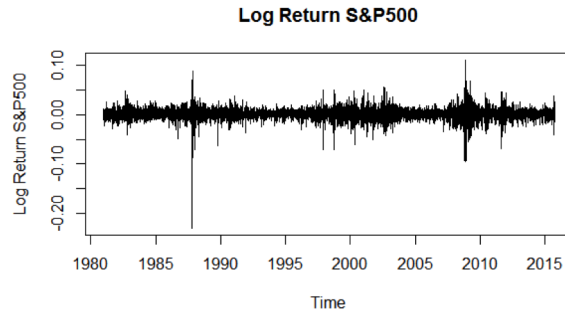
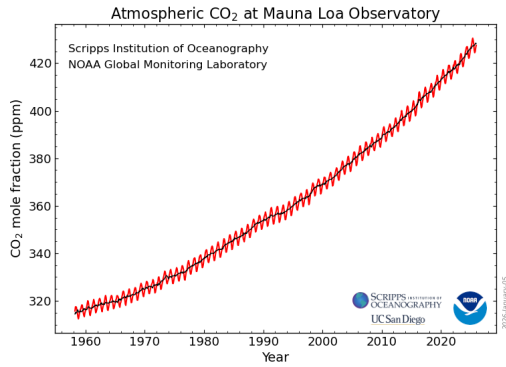
DEF 2.2 We call a time series X_t *strictly stationary*, or that it follows a stationary distribution, if

$$(X_{t_1}, \dots, X_{t_n}) \stackrel{d}{=} (X_{t_1+h}, \dots, X_{t_n+h})$$

DEF 2.3 for any choice of $t \in \mathbb{Z}^n$ and $h \in \mathbb{Z}$, which we call *lag*.

Setting $n = 1$, we see that $X_t \stackrel{d}{=} X_{t'} \forall t \neq t' \in \mathbb{Z}$.

This is akin to the notion of time homogeneity of Markov chains.



The left time series is not strictly stationary, but the right is.

For a time series X_t with $\mathbb{E}[X_t^2] < \infty$, we define

$$\mu(t) = \mathbb{E}[X_t] \quad \gamma(s, t) = \text{cov}(X_s, X_t)$$

DEF 2.4 called the *mean function* and *autocovariance*, respectively.

DEF 2.5 For a time series X_t with $\mathbb{E}[X_t^2] < \infty$, we say that it is *covariance stationary* if

$$\mu(t) = \mu \quad \text{and} \quad \gamma(s, t) = \gamma(s + h, t + h)$$

for integers s, t , and h . In this case, we write $\gamma(h) := \gamma(0, h)$.

PROP 2.1 For a covariance-stationary time series, $\gamma(0) = \text{Var}(X_t)$.

PROP 2.2 For a time series X_t with $\mathbb{E}[X_t^2] < \infty$, if it is strictly stationary it is covariance stationary..

DEF 2.7 Let X_t be covariance stationary. The *autocorrelation* of X_t , also called the serial correlation, is defined to be

$$\rho(h) = \text{cor}(X_0, X_h) = \frac{\text{cov}(X_0, X_h)}{\sqrt{\text{Var}(X_0)\text{Var}(X_h)}} = \frac{\gamma(h)}{\gamma(0)}$$

This definition is consistent with the usual notion of correlation, i.e.

$$\text{cor}(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Shape

In addition to time-dependency, we are also interested in our returns' overall distribution. For instance, we expect heavy tails and right-skewness, as mentioned in [Def 1.14](#).

Let X have mean μ and variance σ^2 . Suppose $\mathbb{E}[|X|^3] < \infty$. Then the *skewness* of X is

DEF 2.8

$$\beta := \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right] = \frac{\mathbb{E}[(X - \mu)^3]}{\mathbb{E}[(X - \mu)^2]^{3/2}}$$

If $\beta < 0$, we say that X is *left skewed*. Similarly, when $\beta > 0$, we say that X is *right skewed*.

DEF 2.9

DEF 2.10

A left skewed distribution typically has its largest mass set to the *right* side, and vice-versa for a right skewed distribution. Despite this, a left skewed distribution indeed has heavier *left* tails, and similarly for right skewed distributions. The naming convention also derives from a left skewed variable's mean being to the left of both its median and mode.

Let X have mean μ and variance σ^2 . Suppose $\mathbb{E}[|X|^4] < \infty$. Then the *kurtosis* of X is

DEF 2.11

$$\kappa := \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] = \frac{\mathbb{E}[(X - \mu)^4]}{\mathbb{E}[(X - \mu)^2]^2}$$

A large kurtosis indicates a higher concentration around the variable's mean and a heavier tail.

A normal variable has skew $\beta = 0$ and kurtosis $\kappa = 3$. We call $\kappa - 3$ the *excess kurtosis* of a variable X .

DEF 2.12

White Noise

It is important that we can capture "randomness" in a model of financial returns.

A *white noise* process X_t , written $X_t \sim \text{WN}(0, \sigma^2)$, is a covariance stationary time series with $\mu(t) = 0$ and $\gamma(t) = 0$.

DEF 2.13

In other words, this is a covariance-stationary time series which shows no autocorrelation.

A *strict white noise* process X_t is any process of IID random variables with finite variance σ^2 . In this case, we write $X_t \sim \text{SWN}(\mu, \sigma^2)$. When $\mu = 0$, a strict white noise is a white noise.

DEF 2.14

A *martingale difference* X_t with respect to $\mathcal{F}_{t-1}$ satisfies $\mathbb{E}[|X_t|] < \infty$ and $\mathbb{E}[X_t | \mathcal{F}_{t-1}] = 0$.

DEF 2.15

The *volatility* of a time series X_t is $\sigma_t = \sqrt{\text{Var}(X_t | \mathcal{F}_{t-1})}$. Similarly, we say that $\mathbb{E}[X_t | \mathcal{F}_{t-1}]$ is the *conditional mean*.

DEF 2.16

DEF 2.17

If X_t is a martingale difference and $\mathbb{E}[X_t^2] < \infty$, then X_t is a white noise process.

PROP 2.3

We need to show that $\mu = 0$ and $\gamma(t) = 0$. We have $\mathbb{E}[X_t] = \mathbb{E}_{\mathcal{F}_{t-1}}[\mathbb{E}[X_t | \mathcal{F}_{t-1}]] = \mathbb{E}_{\mathcal{F}_{t-1}}[0] = 0$. For autocorrelation, with $s < t$, $\text{cov}(X_s, X_t) = \mathbb{E}[X_s X_t]$. Expanding,

PROOF.

$$\mathbb{E}_{\mathcal{F}_{t-1}}[\mathbb{E}[X_s X_t | \mathcal{F}_{t-1}]] = \mathbb{E}_{\mathcal{F}_{t-1}}[X_s \mathbb{E}[X_t | \mathcal{F}_{t-1}]] = X_s \mathbb{E}_{\mathcal{F}_{t-1}}[\mathbb{E}[X_t | \mathcal{F}_{t-1}]] = 0$$

noting that $X_s \in \mathcal{F}_{t-1}$. □

Estimators and Tests

For the following, let $X_1, \dots, X_n$ be time series observations. We are interested in estimating and testing the shape, autocorrelation, and white-noise-ness of empirical data.

DEF 2.18 The *sample skewness* and *sample kurtosis* are given by

$$\hat{\beta}_n := \frac{1/n \sum_{i=1}^n (X_i - \bar{X})^3}{\left[1/n \sum_{i=1}^n (X_i - \bar{X})^2\right]^{3/2}} \quad \hat{\kappa}_n := \frac{1/n \sum_{i=1}^n (X_i - \bar{X})^4}{\left[1/n \sum_{i=1}^n (X_i - \bar{X})^2\right]^2}$$

DEF 2.20 The *sample autocovariance* is given by

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (X_t - \bar{X})(X_{t+h} - \bar{X})$$

DEF 2.21 It follows that the *sample autocorrelations* are given by

$$\hat{\rho}(h) := \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

DEF 2.22 We call the map $h \mapsto \hat{\rho}(h)$ the *correlogram*.

DEF 2.23 The *Jarque-Bera statistic* is

$$T_n = \frac{n}{6} \left(\beta_n^2 + \frac{1}{4} (\kappa_n - 3)^2 \right)$$

Under the null $H_0 : X \sim \mathcal{N}(\mu, \sigma^2)$, where μ and σ are unknown, $T_n \sim \chi_2^2$.

DEF 2.24 The *Ljung-Box statistic* is

$$Q_n = n(n+2) \sum_{j=1}^h \frac{\widehat{\rho}(j)^2}{n-j}$$

Under the null $H_0 : [X_1, \dots, X_n]$ is a strict white noise, $Q_n \sim \chi_h^2$.

Stylized Facts of Financial Time Series

DEF 2.25 A *stylized fact* is an assumption about the empirical behavior of a financial return series. The following lists the 7 stylized facts we concern ourselves with in this course, along with whether certain time series models (which we will discuss later) are capable of reproducing them.

Description	ARMA	GARCH	ARMA-GARCH
X_t should form a white noise, but not a strict white noise.	No	Yes	Yes
$ X_t $ exhibits large autocorrelation.	No	Yes	Yes
$\mathbb{E}[X_t \mathcal{F}_{t-1}] \approx 0$.	Yes	Yes	Yes
Volatility varies over time.	No	Yes	Yes
Extreme returns appear in clusters.	No	Yes	Yes
X_t has excess kurtosis.	No	Yes	Yes

ARMA-GARCH MODELS

We will consider parameterized models for financial time series, which, ideally, satisfy stylized facts and serve as a faithful input for risk measure computations.

ARMA

In the present section, $\varepsilon_t \sim \text{WN}(0, \sigma_\varepsilon^2)$ is a white noise. In the context of time series models, we say that ε_t are *innovations*. DEF 2.26

MOVING AVERAGE PROCESS

The series $X_t \sim \text{MA}(q)$ forms a *moving average process* with lag q if there are coefficients $\theta_i \in \mathbb{R}$ which satisfy DEF 2.27

$$X_t = \sum_{i=1}^q \theta_i \varepsilon_{t-i} + \varepsilon_t$$

We observe autocorrelation up to lag q , and no autocorrelation whatsoever for larger lags.

AUTOREGRESSIVE PROCESS

The series $X_t \sim \text{AR}(p)$ forms an *autoregressive process* with lag p if there are coefficients $\phi_j \in \mathbb{R}$ which satisfy DEF 2.28

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + \varepsilon_t$$

We observe strong autocorrelation up to lag p , which gradually decays at larger lags.

Note that, as the autoregressive process is a solution to a stochastic difference equation, computations must be done iteratively, with an initial condition. In contrast, the moving average process may be computed at once.

Eg. 2.1 We will expand an AR(1) process. We have

$$\begin{aligned} X_t &= \phi X_{t-1} + \varepsilon_t \\ &= \phi(\phi X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \phi^2 X_{t-2} + \phi \varepsilon_{t-1} + \varepsilon_t \\ &= \phi^2(\phi X_{t-3} + \varepsilon_{t-2}) + \phi \varepsilon_{t-1} + \varepsilon_t = \phi^3 X_{t-3} + \phi^2 \varepsilon_{t-2} + \phi \varepsilon_{t-1} + \varepsilon_t \\ &\vdots \\ &= \phi^k X_{t-k} + \sum_{i=0}^{k-1} \phi^i \varepsilon_{t-i} \end{aligned}$$

So $X_t \approx \lim_{k \rightarrow \infty} \phi^k X_{t-k} + \sum_{i=0}^{\infty} \phi^i \varepsilon_{t-i}$. The summation only converges when $|\phi| < 1$, in which case $X_t \approx \sum_{i=0}^{\infty} \phi^i \varepsilon_{t-i}$. In fact, $|\phi| < 1$ is a necessary condition for a solution to exist.

MOVING AVERAGE AUTOREGRESSIVE PROCESS

DEF 2.29 The series $X_t \sim \text{ARMA}(p, q)$ forms an *autoregressive moving average process* with lags p and q if it is covariance-stationary and there are coefficients $\theta_i, \phi_j \in \mathbb{R}$ which satisfy

$$X_t = \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t \quad t \in \mathbb{Z}$$

If X_t is an $\text{ARMA}(p, q)$ process with mean μ , then $X_t - \mu$ is an $\text{ARMA}(p, q)$ process with mean 0. Hence, we frequently write

$$X_t = \mu + \underbrace{\sum_{i=1}^p \phi_i (X_{t-i} - \mu) + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t}_{:= \mu_t} = \mu_t + \mu$$

ARIMA PROCESSES

Sometimes our data is non-stationary and thus cannot be modeled directly by an ARMA process. We can manually remove autocorrelations in the following way:

$$\nabla X_t = X_t - X_{t-1} \implies \nabla^d X_t = \nabla^{d-1}(X_t - X_{t-1})$$

DEF 2.30 We call $X_t \sim \text{ARIMA}(p, d, q)$ an *ARIMA process* if $\nabla^d X_t \sim \text{ARMA}(p, q)$.

CAUSALITY

We consider processes that can be defined explicitly in terms of its innovations. We say that

DEF 2.31 X_t is *causal* if there are coefficients ψ_i with

$$X_t = \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i} \quad \text{with} \quad \sum_{i=0}^{\infty} |\psi_i| < \infty$$

We also consider a mean-shifted $X_t + \mu$ to be causal if X_t is causal.

PROP 2.4 If X_t is causal, then it is covariance-stationary with

$$\text{Var}(X_t) = \sum_{i=0}^{\infty} \psi_i^2 \sigma_\varepsilon^2 \quad \rho(h) = \frac{\sum_{i=0}^{\infty} \psi_i \psi_{i+|h|}}{\sum_{i=0}^{\infty} \psi_i^2}$$

PROOF.

$$\text{Var}(X_t) = \sum_{i=0}^{\infty} \text{Var}(\psi_i \varepsilon_{t-i}) = \sum_{i=0}^{\infty} \psi_i^2 \sigma_\varepsilon^2.$$

Finally, recalling that covariance is bilinear, we compute:

$$\text{cov}(X_t, X_{t+h}) = \text{cov}\left(\sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i}, \sum_{j=0}^{\infty} \psi_j \varepsilon_{t+h-j}\right) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \psi_i \psi_j \text{cov}(\varepsilon_{t-i}, \varepsilon_{t+h-j})$$

but our innovations are white noise, and thus exhibit no autocovariance. Hence,

$\text{cov}(\varepsilon_{t-i}, \varepsilon_{t+h-j}) = 0$ unless $j = i + h$, in which case it is σ_ε^2 . We rearrange, then,

$$\text{cov}(X_t, X_{t+h}) = \sum_{i=0}^{\infty} \psi_i \psi_{i+h} \sigma_\varepsilon^2 = \sum_{i=0}^{\infty} \psi_i \psi_{i+|h|} \sigma_\varepsilon^2$$

For the last equivalence, suppose $h < 0$ (in which case $h \neq |h|$). Then ψ_{i+h} doesn't make sense for $i < |h|$. Thus:

$$\text{cov}(X_t, X_{t+h}) = \sum_{i=|h|}^{\infty} \psi_i \psi_{i+h} \sigma_\varepsilon^2 = \sum_{j=0}^{\infty} \psi_{j+|h|} \psi_j \sigma_\varepsilon^2 \text{ after re-indexing } j = i + |h|$$

Since our equation for $\gamma(t, t+h) = \text{cov}(X_t, X_{t+h})$ does not rely on t , we conclude that X_t is covariance-stationary, and write $\gamma(t, t+h) = \gamma(h)$. Diving $\frac{\gamma(h)}{\gamma(0)} = \frac{\gamma(h)}{\text{Var}(X_t)} = \rho(h)$ gives the autocorrelation result. $\square$

Theorem 2.1 Causality of an ARMA Process

Let $X_t \sim \text{ARMA}(p, q)$. Then X_t is casual, strictly stationary, and unique if the complex polynomials

$$\tilde{\phi}(z) = 1 - \sum_{i=1}^p \phi_i z^i \quad \tilde{\theta}(z) = 1 + \sum_{j=1}^q \theta_j z^j$$

have no common factors, and if $\tilde{\phi}(z)$ has no complex roots for $|z| \leq 1$. In this case, our coefficients are determined by solving

$$\sum_{i=0}^{\infty} \psi_i z^i = \frac{\tilde{\theta}(z)}{\tilde{\phi}(z)} : |z| \leq 1$$

Eg. 2.2 [Example 2.1](#) establishes that AR(1) is causal whenever $|\phi| < 1$. It's easy to see that any moving average process MA(q) is also causal. This allows us to compute their autocorrelations easily using the equations in [Prop 2.4](#). In particular, an ARMA(1, 1) process has autocorrelation

$$\rho_{\text{ARMA}(1,1)}(h) = \frac{\phi_1^{|h|-1} (\theta_1 + \phi_1)(1 + \theta_1 \phi_1)}{\theta_1^2 + 2\theta_1 \phi_1 + 1}$$

Eg. 2.3 MA(1) is causal by definition. To come to this conclusion from the theory above, we have

$$\tilde{\phi}(z) = 1 \quad \tilde{\theta}(z) = 1 + \theta z$$

Clearly $\tilde{\phi}(z)$ does not have roots. We have

$$\sum_{i=0}^{\infty} \psi_i z^i = \frac{\tilde{\theta}(z)}{\tilde{\phi}(z)} = 1 + \theta z$$

Eg. 2.4 AR(1) is causal, as discussed, when $|\phi| < 1$. To come at this conclusion from the theory above, we have

$$\tilde{\phi}(z) = 1 - \theta z \quad \tilde{\theta}(z) = 1$$

In order for $1 - \phi z$ to have no roots in the unit circle, we require that $|\frac{1}{\phi}| > 1 \implies |\phi| < 1$, as found before.

INVERTIBILITY

Conversely, we consider processes whose innovations can be written explicitly in terms of past series term. In this case, ε_t is measurable with respect to $\mathcal{F}_{t-1}$. In particular, X_t is

DEF 2.32 *invertible* if

$$\varepsilon_t = \sum_{i=0}^{\infty} \zeta_i X_{t-i} \quad \text{with} \quad \sum_{i=0}^{\infty} |\zeta_i| < \infty$$

Theorem 2.2 Invertibility of an ARMA Process

Let $X_t \sim \text{ARMA}(p, q)$. Then X_t is invertible if the complex polynomial

$$\tilde{\theta}(z) = 1 + \sum_{j=1}^q \theta_j z^j$$

has no complex roots for $|z| < 1$. For the case ARMA(1, 1), this condition reduces to $|\theta_1| < 1$.

PROP 2.5 Let X_t be a causal, invertible ARMA process with mean μ . Then

1. When ε_t are martingale differences, $\mathbb{E}[X_t | \mathcal{F}_{t-1}] = \mu_t$.
2. When $\varepsilon_t \sim \text{SWN}(0, \sigma_\varepsilon^2)$, $\text{Var}(X_t | \mathcal{F}_{t-1}) = \sigma_\varepsilon^2$.

PROOF.

Let ε_t be a martingale difference. Then $\mathbb{E}[X_t | \mathcal{F}_{t-1}] = \mathbb{E}[\mu_t + \varepsilon_t | \mathcal{F}_{t-1}] = \mathbb{E}[\mu_t | \mathcal{F}_{t-1}]$. But μ_t depends entirely on $\mathcal{F}_{t-1}$, so $\mathbb{E}[\mu_t] = \mu_t$. Similarly, if $\varepsilon_t \sim \text{SWN}(0, \sigma_\varepsilon^2)$, then $\text{Var}(\mu_t + \varepsilon_t | \mathcal{F}_{t-1}) = \text{Var}(\varepsilon_t | \mathcal{F}_{t-1}) = \text{Var}(\varepsilon_t) = \sigma_\varepsilon^2$, by independence. $\square$

♣ From now on, all ARMA processes are assumed to be causal and invertible ♠

GARCH

Classical ARMA models, like those shown above, do not capture changing volatility, which markets generally exhibit. GARCH processes are design to capture changing volatility.

Let $X_t = \sigma_t Z_t$, where $Z_t \sim \text{SWN}(0, 1)$. Define

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2 \quad \alpha_0 > 0 \quad \alpha_i, \beta_j \geq 0$$

Typically, $Z_t \sim \mathcal{N}(0, 1)$ or a scaled t_ν .

If X_t is a strictly stationary solution to the equations above, we call $X_t \sim \text{GARCH}(p, q)$ a **GARCH process** with parameters p and q . Sometimes we write $\text{ARCH}(p) = \text{GARCH}(p, 0)$.

DEF 2.33

Let $X_t \sim \text{GARCH}(p, q)$. Then $\text{Var}(X_t) = \mathbb{E}[X_t^2] = \mathbb{E}[\sigma_t^2]$.

PROP 2.6

We have $\text{Var}(X_t) = \text{Var}(\sigma_t Z_t) = \mathbb{E}[\sigma_t^2 Z_t^2] - \mathbb{E}[\sigma_t Z_t]^2 = \mathbb{E}[\sigma_t^2 Z_t^2] = \mathbb{E}[\sigma_t^2]$. $\square$

PROOF.

Let $X_t \sim \text{GARCH}(p, q)$ process. Then, if $\mathbb{E}[X_t^2] < \infty$,

PROP 2.7

$$\mathbb{E}[X_t | \mathcal{F}_{t-1}] = 0 \quad \sqrt{\text{Var}(X_t | \mathcal{F}_{t-1})} = \sigma_t$$

In short, X_t is a martingale difference and white noise, with volatility σ_t .

We have

$$\mathbb{E}[X_t | \mathcal{F}_{t-1}] = \mathbb{E}[\sigma_t Z_t | \mathcal{F}_{t-1}] = \mathbb{E}[Z_t] \mathbb{E}[\sigma_t | \mathcal{F}_{t-1}] = 0$$

since Z_t is IID with mean 0. Similarly,

$$\text{Var}(X_t | \mathcal{F}_{t-1}) = \text{Var}(\sigma_t Z_t | \mathcal{F}_{t-1}) = \mathbb{E}[\sigma_t^2 Z_t^2 | \mathcal{F}_{t-1}] - \mathbb{E}[\sigma_t Z_t | \mathcal{F}_{t-1}]^2 = \mathbb{E}[\sigma_t^2 Z_t^2 | \mathcal{F}_{t-1}]$$

Then, by independence, and since $\mathbb{E}[Z_t^2] = \text{Var}(Z_t) = 1$, we have $\mathbb{E}[\sigma_t^2 Z_t^2 | \mathcal{F}_{t-1}] = \mathbb{E}[\sigma_t^2 | \mathcal{F}_{t-1}]$. But σ_t^2 is measurable w.r.t. $\mathcal{F}_{t-1}$, so $\mathbb{E}[\sigma_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2$, as desired.

Since $\mathbb{E}[X_t^2] < \infty$ and X_t is strictly stationary, note also that X_t covariance-stationary. Since X_t is a martingale difference, it is also a white noise, by [Prop 2.3](#). $\square$

PROOF.

Eg. 2.5 A $\text{GARCH}(1, 1)$ process has

$$X_t = \sigma_t Z_t \quad \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2 = \alpha_0 + \sigma_{t-1}^2 (\alpha_1 Z_{t-1}^2 + \beta_1)$$

Set $Y_t = \sigma_t^2$ and $B_t = \alpha_0$ (a constant variable) and $A_t = \alpha_1 Z_{t-1}^2$. Then, σ_t^2 satisfies the stochastic recurrence equation

$$Y_t = A_t Y_{t-1} + B_t : t \in \mathbb{Z}$$

The theory of stochastic processes provides that there is a unique strictly stationary solution Y_t if $\mathbb{E}[\max(0, \log |B_t|)] < \infty$ and $\mathbb{E}[\log |A_t|] < 0$. The first condition holds trivially. The second provides the basis for the theorem below.

Theorem 2.3 Nelson's Theorem

A unique, stationary solution for GARCH(1, 1) exists if and only if

$$\mathbb{E}[\log(\alpha_1 Z_t^2 + \beta_1)] < 0$$

Moreover, the solution X_t can be written as

$$\sigma_t^2 = \alpha_0 \left(1 + \sum_{i=1}^{\infty} \prod_{j=1}^i (\alpha_1 Z_{t-j}^2 + \beta_1) \right) \quad X_t = Z_t \sqrt{\alpha_0 \left(1 + \sum_{i=1}^{\infty} \prod_{j=1}^i (\alpha_1 Z_{t-j}^2 + \beta_1) \right)}$$

PROOF.

The first statement is taken as black-box from the theory of stochastic recurrences. Let X_t be a stationary solution. Then

$$\begin{aligned} \sigma_t^2 &= \alpha_0 + \sigma_{t-1}^2 (\alpha_1 Z_{t-1}^2 + \beta_1) \\ &= \alpha_0 + (\alpha_0 + \sigma_{t-2}^2 (\alpha_1 Z_{t-2}^2 + \beta_1)) (\alpha_1 Z_{t-1}^2 + \beta_1) \\ &\vdots \\ &= \alpha_0 + \sum_{i=1}^{\infty} \alpha_0 \prod_{j=0}^{i-1} (\alpha_1 Z_{t-j-1}^2 + \beta_1) = \alpha_0 + \sum_{i=1}^{\infty} \alpha_0 \prod_{j=1}^i (\alpha_1 Z_{t-j-1}^2 + \beta_1) \\ &= \alpha_0 \left[1 + \sum_{i=1}^{\infty} \prod_{j=0}^{i-1} (\alpha_1 Z_{t-j-1}^2 + \beta_1) \right] \end{aligned}$$

Recalling $X_t = \sigma_t Z_t$ gives the result. $\square$

Theorem 2.4 Bollerslev's Theorem

A GARCH(1, 1) process is covariance-stationary if and only if

$$\alpha_1 + \beta_1 < 1$$

In this case, the unconditional variance of X_t is given by

$$\text{Var}(X_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Generally, a GARCH(p, q) process is covariance-stationary if and only if

$$\sum_{i=1}^p \alpha_i + \sum_{j=1}^q \beta_j < 1$$

in which case $\text{Var}(X_t) = \frac{\alpha_0}{1 - \sum_{i=1}^p \alpha_i - \sum_{j=1}^q \beta_j}$.

PROOF.

We'll give a treatment of the GARCH(1, 1) case.

($\Rightarrow$) Suppose X_t is a covariance-stationary solution. We have

$$\infty > \mathbb{E}[X_t^2] = \mathbb{E}[\sigma_t^2 Z_t^2] = \mathbb{E}[\sigma_t^2] \mathbb{E}[Z_t^2] = \mathbb{E}[\sigma_t^2]$$

We compute $\mathbb{E}[\sigma_t^2]$:

$$\begin{aligned} \mathbb{E}[\sigma_t^2] &= \mathbb{E}[\alpha_0 + \sigma_{t-1}^2(\alpha_1 Z_{t-1}^2 + \beta_1)] \\ &= \mathbb{E}[\alpha_0 + (\alpha_0 + \sigma_{t-2}^2(\alpha_1 Z_{t-2}^2 + \beta_1))(\alpha_1 Z_{t-1}^2 + \beta_1)] \\ &= \alpha_0 + \mathbb{E}[\sigma_{t-1}^2](\alpha_1 + \beta_1) \end{aligned}$$

Since X_t is covariance-stationary, $\mathbb{E}[X_t^2] = \mathbb{E}[\sigma_t^2]$ is constant over t . Rearranging, then

$$\mathbb{E}[\sigma_t^2] = \frac{\alpha_0}{1 - \alpha_1 - \beta_1} \geq 0 \implies \alpha_1 + \beta_1 < 1$$

($\Leftarrow$) On the other hand, suppose $\alpha_1 + \beta_1 < 1$. Then,

$$0 > \log(\alpha_1 + \alpha_2) = \log(\mathbb{E}[\alpha_1 Z_t^2 + \beta_1]) \geq \mathbb{E}[\log(\alpha_1 Z_t^2 + \beta_1)]$$

by Jensen's inequality. It follows by the previous theorem that some strictly stationary solution exists. It remains to show $\mathbb{E}[X_t^2] = \mathbb{E}[\sigma_t^2] < \infty$. One writes

$$\mathbb{E}[\sigma_t^2] = \mathbb{E}\left[\alpha_0 + \sum_{i=1}^{\infty} \alpha_0 \prod_{j=1}^i (\alpha_1 Z_{t-j}^2 + \beta_1)\right] = \alpha_0 + \sum_{i=1}^{\infty} \alpha_0 \mathbb{E}\left[\prod_{j=1}^i (\alpha_1 Z_{t-j}^2 + \beta_1)\right]$$

By independence of the Z_t 's, we get

$$\mathbb{E}[\sigma_t^2] = \alpha_0 + \sum_{i=1}^{\infty} \alpha_0 \prod_{j=1}^i (\alpha_1 + \beta_1) = \alpha_0 + \alpha_0 \sum_{i=1}^{\infty} (\alpha_1 + \beta_1)^i = \frac{\alpha_0}{1 - \alpha_1 - \beta_1} < \infty$$

as desired. Note that $\text{Var}(X_t) = \mathbb{E}[X_t^2] = \mathbb{E}[\sigma_t^2]$. □

For a GARCH(1, 1) process, $\mathbb{E}[X_t^{2m}] < \infty \iff \mathbb{E}[Z_t^{2m}] < \infty$ and $\mathbb{E}[(\alpha_1 Z_t^2 + \beta_1)^m] < 1$. In particular, the kurtosis κ_{X_t} exists if and only if $\mathbb{E}[X_t^4] = \mathbb{E}[\sigma_t^4] \mathbb{E}[Z_t^4] = \mathbb{E}[\sigma_t^4] \kappa_Z < \infty$, i.e.

PROP 2.8

$$\kappa_Z < \infty \text{ and } \alpha_1 + \beta_1 < \sqrt{1 - \alpha_1^2(\kappa_Z - 1)}$$

We have that $\mathbb{E}[X_t^{2m}] < \infty \iff \mathbb{E}[Z_t^{2m}] < \infty$ and $\mathbb{E}[\sigma_t^{2m}] < \infty$. One can prove that $\mathbb{E}[\sigma_t^{2m}] < \infty \iff \mathbb{E}[(\alpha_1 Z_t^2 + \beta_1)^m] < 1$.

PROOF.

Adapting this, $\mathbb{E}[X_t^4] < \infty \iff \mathbb{E}[Z_t^4] < \infty$ and $\mathbb{E}[(\alpha_1 Z_t^2 + \beta_1)^2] < 1$. Note that, since

$\mathbb{E}[Z_t] = 0$ and $\text{Var}(Z_t) = 1$, $\mathbb{E}[Z_t^4] = \kappa_Z$ is its kurtosis. Then

$$\begin{aligned}\mathbb{E}[(\alpha_1 Z_t^2 + \beta_1)^2] &= \mathbb{E}[\alpha_1^2 Z_t^4 + \beta_1^2 + 2\beta_1 \alpha_1 Z_t^2] \\ &= (\alpha_1 + \beta_1)^2 + \alpha_1^2(\kappa_Z - 1) < 1 \\ \iff \alpha_1 + \beta_1 &< \sqrt{1 - \alpha_1^2(\kappa_Z - 1)}\end{aligned}$$

PROP 2.9 If a GARCH(1, 1) process has finite kurtosis κ_{X_t} , then

$$\kappa_{X_t} = \kappa_Z \frac{1 - (\alpha_1 + \beta_1)^2}{1 - (\alpha_1 + \beta_1)^2 - \alpha_1(\kappa_Z - 1)}$$

PROOF.

$\mathbb{E}[X_t^4] = \kappa_Z \mathbb{E}[\sigma_t^4]$. Then,

$$\begin{aligned}\sigma_t^4 &= (\alpha_0 + \sigma_{t-1}^2(\alpha_1 Z_{t-1}^2 + \beta_1))^2 \\ &= \alpha_0^2 + \sigma_{t-1}^4(\alpha_1 Z_{t-1}^2 + \beta_1)^2 + 2\alpha_0 \sigma_{t-1}^2(\alpha_1 Z_{t-1}^2 + \beta_1) \\ &= \alpha_0^2 + \sigma_{t-1}^4(\alpha_1^2 Z_{t-1}^4 + \beta_1^2 + 2\alpha_1 \beta_1 Z_{t-1}^2) + 2\alpha_0 \sigma_{t-1}^2(\alpha_1 Z_{t-1}^2 + \beta_1)\end{aligned}$$

In expectation, then

$$\mathbb{E}[\sigma_t^4] = \alpha_0^2 + \mathbb{E}[\sigma_t^4](\alpha_1^2 \kappa_Z + \beta_1^2 + 2\alpha_1 \beta_1) + 2\alpha_0 \mathbb{E}[\sigma_t^2](\alpha_1 + \beta_1)$$

Rearranging for $\mathbb{E}[\sigma_t^4]$, and recalling $\mathbb{E}[\sigma_t^2] = \mathbb{E}[X_t^2]$, we get

$$\mathbb{E}[\sigma_t^4] = \frac{\alpha_0^2 + 2\alpha_0(\alpha_1 + \beta_1)\mathbb{E}[X_t^2]}{1 - \alpha_1^2(\kappa_Z - 1) - (\alpha_1 + \beta_1)^2}$$

Previously, we've calculated that $\mathbb{E}[X_t^2] = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$. Then,

$$\begin{aligned}\kappa_{X_t} &= \frac{\mathbb{E}[X_t^4]}{\mathbb{E}[X_t^2]^2} = \kappa_Z \frac{\alpha_0^2 + 2\alpha_0(\alpha_1 + \beta_1)\mathbb{E}[X_t^2]}{1 - \alpha_1^2(\kappa_Z - 1) - (\alpha_1 + \beta_1)^2} \frac{1}{\mathbb{E}[X_t^2]^2} \\ &= \kappa_Z \frac{(1 - \alpha_1 - \beta_1)^2 + 2(\alpha_1 + \beta_1)(1 - \alpha_1 - \beta_1)}{1 - \alpha_1^2(\kappa_Z - 1) - (\alpha_1 + \beta_1)^2} \\ &= \kappa_Z \frac{1 - (\alpha_1 + \beta_1)^2}{1 - \alpha_1^2(\kappa_Z - 1) - (\alpha_1 + \beta_1)^2}\end{aligned}$$

as desired. $\square$

LEVERAGED GARCH

Sometimes it's reasonable to assume that bad returns will have a larger effect on volatility.

We can adapt our series to this assumption by writing, for a GARCH(1, 1) process,

$$\sigma_t^2 = \alpha_0 + \alpha_1(X_{t-1} - \delta|X_{t-1}|)^2 + \beta_1 \sigma_{t-1}^2 = \begin{cases} \alpha_0 + \alpha_1(1 - \delta)^2 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2 & X_{t-1} \geq 0 \\ \alpha_0 + \alpha_1(1 + \delta)^2 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2 & X_{t-1} < 0 \end{cases}$$

where, typically, $\delta \in (0, 1)$ is our *leverage parameter*. Note that, when $\delta < 1$, we get leverage toward positive returns. which is a typically positive. Asymmetry can also be captured by asymmetric (e.g. skewed) innovation distributions.

DEF 2.34

ARMA-GARCH

Let $X_t = \mu_t + \varepsilon_t$ be an $\text{ARMA}(p_1, q_1)$ process with mean μ . Suppose that $\varepsilon_t \sim \text{GARCH}(p_2, q_2)$. Write

As opposed to
 $\text{WN}(0, \sigma_\varepsilon^2)$

$$X_t = \mu_t + \varepsilon_t \text{ with } \begin{cases} \mu_t = \mu + \sum_{i=1}^{p_1} \phi_i (X_{t-i} - \mu) + \sum_{j=1}^{q_1} \theta_j (X_{t-j} - \mu_{t-j}) \\ \varepsilon_t = \sigma_t Z_t \\ \sigma_t^2 = \alpha_0 + \sum_{i=1}^{p_2} \alpha_i (X_{t-i} - \mu_{t-i})^2 + \sum_{j=1}^{q_2} \beta_j \sigma_{t-j}^2 \\ Z_t \sim \text{SWN}(0, 1) \end{cases}$$

If $X_t \sim \text{ARMA}(p_1, q_1) - \text{GARCH}(p_2, q_2)$ satisfies the above process and ε_t is covariance stationary, then we call it an *ARMA(p_1, q_1) - GARCH(p_2, q_2) process*. We still refer to ε_t as "innovations," regardless of the fact that they are dictated by their own GARCH process.

DEF 2.35

If X_t is an ARMA-GARCH process, then

PROP 2.10

$$\mathbb{E}[X_t | \mathcal{F}_{t-1}] = \mu_t \quad \text{and} \quad \text{Var}(X_t | \mathcal{F}_{t-1}) = \sigma_t^2$$

Similarly, for a lag $h \geq 0$, $\mathbb{E}[X_{t+h} | \mathcal{F}_t] = \mathbb{E}[\mu_{t+h} | \mathcal{F}_t]$ and $\text{Var}(X_{t+h} | \mathcal{F}_t) = \mathbb{E}[\sigma_{t+h}^2 | \mathcal{F}_t]$.

We'll prove the first statement. The lagged result follows a similar argument.

PROOF.

$$\mathbb{E}[X_t | \mathcal{F}_{t-1}] = \mathbb{E}[\mu_t | \mathcal{F}_{t-1}] + \mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] = \mu_t.$$

Similarly, $\text{Var}(X_t | \mathcal{F}_{t-1}) = \text{Var}(\mu_t | \mathcal{F}_{t-1}) + \text{Var}(\varepsilon_t | \mathcal{F}_{t-1}) = \text{Var}(\sigma_t Z_t | \mathcal{F}_{t-1}) = \sigma_t^2 \text{Var}(Z_t | \mathcal{F}_{t-1})$, since σ_t^2 determined completely by $\mathcal{F}_{t-1}$. Then, $\text{Var}(Z_t | \mathcal{F}_{t-1}) = 1$ gives the result. $\square$

FITTING TIME-SERIES MODELS

Conditional Likelihood

In this section, we wish to train a model on realized data $x_1, \dots, x_n$, with possible access to data $x_s : s \leq 0$. To fit an ARMA, GARCH, or ARMA-GARCH model to this data, we maximize the conditional likelihood of the data. In particular, let $f_\Theta(x_1, \dots, x_n | \mathcal{F}_0)$ be the joint conditional density of data $x_1, \dots, x_n$ with respect to $\mathcal{F}_0$. Then the *conditional likelihood* of $x_1, \dots, x_n$ with respect to $\mathcal{F}_0$ and parameter set Θ is

DEF 2.36

$$\begin{aligned} \mathcal{L}_{\Theta | \mathcal{F}_0}(x_1, \dots, x_n) &= f_\Theta(x_1, \dots, x_n | \mathcal{F}_0) = f_\Theta(x_1 | \mathcal{F}_0) f_\Theta(x_2 | x_1, \mathcal{F}_0) \cdot \dots \cdot f_\Theta(x_n | x_n, \dots, x_1, \mathcal{F}_0) \\ &= \prod_{t=1}^n f_\Theta(x_t | x_{t-1}, \dots, x_1, \mathcal{F}_0) \\ \implies \log(\mathcal{L}_{\Theta | \mathcal{F}_0}(x_1, \dots, x_n)) &= \sum_{t=1}^n \log(f_\Theta(x_t | x_{t-1}, \dots, x_1, \mathcal{F}_0)) \end{aligned}$$

PROP 2.11 Let $X_t = \mu_t + \varepsilon_t$ be an ARMA-GARCH process. If Z_t has a CDF F_Z and density f_Z , then the conditional distribution $(X_t|X_{t-1}, \dots, X_1, \mathcal{F}_0)$ has CDF $F_Z(\frac{x-\mu_t}{\sigma_t})$ and density $f_Z(\frac{x-\mu_t}{\sigma_t}) \frac{1}{\sigma_t}$.

PROOF.

μ_t, σ_t are measurable with respect to $X_{t-1}, \dots, X_1, \mathcal{F}_0$, so we treat them as constants.

$$\begin{aligned} \mathbb{P}(X_t \leq x|X_{t-1}, \dots, X_1, \mathcal{F}_0) &= \mathbb{P}(\mu_t + \sigma_t Z_t \leq x|X_{t-1}, \dots, X_1, \mathcal{F}_0) \\ &= \mathbb{P}\left(Z_t \leq \frac{x - \mu_t}{\sigma_t} | X_{t-1}, \dots, X_1, \mathcal{F}_0\right) = F_Z\left(\frac{x - \mu_t}{\sigma_t}\right) \end{aligned}$$

Differentiating with respect to x , we get the stated density.

PROP 2.12 We do not need all of $X_{t-1}, \dots, X_1, \mathcal{F}_0$ to measure X_t . Consider the following specializations:

Model	$(X_t X_{t-1}, \dots, X_1, \mathcal{F}_0)$	Initial History	Justification
ARCH(1)	$(X_t X_{t-1})$	x_0	$X_t = \alpha_0 + \alpha_1 X_{t-1}^2$
GARCH(1, 1)	$(X_t X_{t-1}, \sigma_{t-1})$	x_0, σ_0	$X_t = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2$
ARMA(1, 1)-GARCH(1, 1)	$(X_t X_{t-1}, \mu_{t-1}, \sigma_{t-1})$	x_0, μ_0, σ_0	see below

PROOF.

For the last justification, we write $X_t = \mu_t + \varepsilon_t$, where

$$\begin{aligned} \mu_t &= \mu + \phi_1(X_{t-1} - \mu) + \theta_1 \varepsilon_{t-1} & \sigma_t^2 &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &= \mu + \phi_1(X_{t-1} - \mu) + \theta_1(X_{t-1} - \mu_{t-1}) & &= \alpha_0 + \alpha_1(X_{t-1} - \mu_{t-1})^2 + \beta_1 \sigma_{t-1}^2 \end{aligned}$$

We treat $\mu \in \Theta$ as a parameter, so this gives the desired result. $\square$

Eg. 2.6 For ARCH(1), we have

$$\mathcal{L}_{\alpha_0, \alpha_1, \mathcal{F}_0} = \mathcal{L}_{\alpha_0, \alpha_1 | x_0} \prod_{t=1}^n \frac{1}{\sqrt{\alpha_0 + \alpha_1 x_{t-1}^2}} f_Z\left(\frac{x_t}{\sqrt{\alpha_0 + \alpha_1 x_{t-1}^2}}\right)$$

Eg. 2.7 For GARCH(1, 1), with $X_t = \sigma_t Z_t$, we have

$$\mathcal{L}_{\alpha_0, \alpha_1, \beta_1 | \mathcal{F}_0} = \mathcal{L}_{\alpha_0, \alpha_1, \beta_1 | x_0, \sigma_0} = \prod_{t=1}^n \frac{1}{\sigma_t} f_Z\left(\frac{x_t}{\sigma_t}\right)$$

where we reconstruct σ_t using x_{t-1} and σ_{t-1} , iteratively, with x_0, σ_0 serving as the base case. In particular, $\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \beta_1 \sigma_{t-1}^2$.

Eg. 2.8 For ARMA(1, 1)-GARCH(1, 1), with $X_t = \mu_t + \varepsilon_t = \mu_t + \sigma_t Z_t$, we have

$$\mathcal{L}_{\theta_1, \phi_1, \alpha_0, \alpha_1, \beta_1 | \mathcal{F}_0} = \mathcal{L}_{\theta_1, \phi_1, \alpha_0, \alpha_1, \beta_1 | x_0, \mu_0, \sigma_0} = \prod_{t=1}^n \frac{1}{\sigma_t} f_Z\left(\frac{x_t - \mu_t}{\sigma_t}\right)$$

where we reconstruct μ_t and σ_t using

$$\mu_t = \mu + \phi_1(X_{t-1} - \mu) + \theta_1 X_{t-1} - \mu_{t-1} \quad \text{and} \quad \sigma_t = \alpha_0 + \alpha_1(X_{t-1} - \mu_{t-1})^2 + \beta_1 \sigma_{t-1}^2$$

Eg. 2.9 For GARCH(p, q), one finds $(X_t | X_{t-1}, \dots, X_1, \mathcal{F}_0) = (X_t | X_{t-1}, \dots, X_{t-p}, \sigma_{t-1}, \dots, \sigma_{t-q})$. Thus, our initial data is $x_0, \dots, x_{p-1}, \sigma_0, \dots, \sigma_{1-q}$, and we recurse to find the conditional likelihood.

Theorem 2.5 Convergence of Conditional MLE on GARCH Models

Let $X_t \sim \text{GARCH}(p, q)$ have true parameters Θ . Suppose we fit a GARCH(p, q) model with $Z \sim \mathcal{N}(0, 1)$ and data $X_1, \dots, X_n$, yielding a parameter set $\hat{\Theta}_n$. Then,

$$\sqrt{n}(\hat{\Theta}_n - \Theta) \rightsquigarrow \mathcal{N}_{p+q+1}(0, J(\Theta)^{-1} I(\Theta) J(\Theta)^{-1})$$

where I and J are (differing forms) of the Fisher information matrix. In particular, if X_t has true innovation distribution $\mathcal{N}(0, 1)$, then $I \equiv J$, and $I \neq J$ otherwise. We call $J^{-1} I J^{-1}$ the *sandwich estimator*.

DEF 2.37

$\hat{\Theta}_n$ is a consistent, asymptotically normal, and asymptotically unbiased estimators of Θ , regardless of whether the model's innovations are correctly specified to be normal.

PROP 2.13

Forecasting

Suppose $X_t \sim \text{ARMA}(p_1, q_1) - \text{GARCH}(p_2, q_2)$. Then, given some $h \geq 0$, the $\mathcal{F}_t$ -measurable variable $\hat{\mu}_{t+h}$ which minimizes $\mathbb{E}[(X_{t+h} - \hat{\mu}_{t+h})^2]$ is exactly $\hat{\mu}_{t+h} = \mathbb{E}[X_{t+h} | \mathcal{F}_t] = \mathbb{E}[\mu_{t+h} | \mathcal{F}_t]$. We call this the *conditional mean prediction* of X_{t+h} .

DEF 2.38

When $X_t \sim \text{ARMA}(1, 1) - \text{GARCH}(1, 1)$ is fitted with parameter set $\hat{\Theta}$, we have

PROP 2.14

$$\hat{\mu}_{t+h} = \mathbb{E}[\mu_{t+h} | \mathcal{F}_t] = \hat{\mu} + \hat{\phi}_1^h (X_t - \hat{\mu}) + \hat{\phi}_1^{h-1} \hat{\theta}_1 \varepsilon_t$$

Similarly, for $h \geq 0$, we call $\hat{\sigma}_{t+h}^2 = \text{Var}(X_{t+h} | \mathcal{F}_{t-1}) = \mathbb{E}[\sigma_{t+h}^2 | \mathcal{F}_t]$ the *volatility forecast*.

DEF 2.39

When $X_t \sim \text{ARMA}(1, 1) - \text{GARCH}(1, 1)$ is fitted with parameter set $\hat{\Theta}$, we have

PROP 2.15

$$\hat{\sigma}_{t+h}^2 = \mathbb{E}(\sigma_{t+h}^2 | \mathcal{F}_t) = \alpha_0 \sum_{i=0}^{h-1} (\alpha_1 + \beta_1)^i + (\alpha_1 + \beta_1)^{h-1} (\alpha_1 X_t^2 + \beta_1 \sigma_t^2)$$

$\hat{\mu}_{t+1}$ and $\hat{\sigma}_{t+1}$ are called *in-sample estimates* when X_{t+1} is in the training data for $\hat{\Theta}$.

DEF 2.40

Model Validation

Suppose a series $X_t \sim \text{ARMA}(p_1, q_1) - \text{GARCH}(p_2, q_2)$ is fitted to the parameter set $\hat{\Theta}$. Then, $\varepsilon_t := X_t - \hat{\mu}_t$ are called the *unstandardized residuals*, where $\hat{\mu}_t$ is the conditional mean prediction of X_t . Furthermore, $\hat{Z}_t := \frac{\varepsilon_t}{\hat{\sigma}_t}$ are called the *standardized residuals*.

DEF 2.41

DEF 2.42

i.e. we correctly specify p and q , but not necessarily the innovation distribution

It should be that $\hat{Z}_t$ approximately follows a SWN distribution, which we can validate using statistical tests (see [Def 2.23](#) and [Def 2.24](#)).

DEF 2.43

To compare two model fits, we primarily use the likelihood ratio test and Akaike information criterion (AIC). In the *likelihood ratio test*, for a complex model C and a basic model $B \subset C$, we measure

$$\text{LRT} = 2(\log(\mathcal{L}_C(x_1, \dots, x_n)) - \log(\mathcal{L}_B(x_1, \dots, x_n)))$$

under the null hypothesis H_0 : [B is a reasonable simplification of C], we have

$$\text{LRT} \sim \chi^2_{|\hat{\Theta}_C| - |\hat{\Theta}_B|}$$

where $|\hat{\Theta}_C|$ are the number of parameters in the model C (and similarly for B). If we reject H_0 , then C is a reasonable extension of B .

Alternatively, we use the AIC index, defined to be

$$2(\dim(C) - \mathcal{L}_C(\vec{x}))$$

where lower is better.

Model-Based Risk Measures

We let VaR_α^t and ES_α^t denote the value-at-risk and expected shortfall of $(X_{t+1}|x_t, \dots, x_1, \mathcal{F}_0) = (X_{t+1}|\mathcal{F}_t)$. In this way, we have that

$$\text{VaR}_\alpha^t = \mu_{t+1} + \sigma_{t+1} \text{VaR}_\alpha(Z_{t+1}) \quad \text{and} \quad \text{ES}_\alpha^t = \mu_{t+1} + \sigma_{t+1} \text{ES}_\alpha(Z_{t+1})$$

When we have a trained model, we use $\hat{\mu}_{t+1}$ and $\hat{\sigma}_{t+1}$ in substitution of the "true" values.

We have some leeway in computing $\text{VaR}_\alpha(Z_{t+1})$ and $\text{ES}_\alpha(Z_{t+1})$. We could either analyze the distribution of standardized residuals $\hat{Z}_t, \hat{Z}_{t-1}, \dots, \hat{Z}_1$, or we could analyze the hypothesized (e.g. normal) distribution directly, in which case we may come to analytical.

Analyzing standardized residuals directly is difficult when α is close to 1 or when we have limited training data. To "extrapolate into the tail," we turn to extreme value theory.

III Extreme Value Theory

We remind ourselves of some results from probability, which we treat as definitions:

1. If $\mathbb{E}[X] < \infty$, then $\bar{X}_n \rightarrow \mathbb{E}[X]$ in probability, i.e. $\lim \mathbb{P}(|\mathbb{E}[X] - \bar{X}_n| > \varepsilon) = 0$ for any $\varepsilon > 0$. This is the *Law of Large Numbers*. DEF 3.1

2. If $\mathbb{E}[X^2] < \infty$, then $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \rightsquigarrow \mathcal{N}(0, 1)$ in distribution. Recall that $A \rightsquigarrow B$ if $\mathbb{P}(A \leq x) \rightarrow \mathbb{P}(B \leq x)$ for all continuity points x . This is the *Central Limit Theorem*. There are a few ways to rearrange this statement. One we'll use frequently is $\frac{S_n - n\mu}{\sqrt{n}\sigma} \rightsquigarrow \mathcal{N}(0, 1)$. DEF 3.2

LIMITING BEHAVIOR OF MAXIMA

Consider n samples of data X_t from a common distribution F . The *maxima*, or worst-case loss, of this sample is $M_n = \max_{i \in [n]} X_i$. Observe that M_n has distribution F^n when samples are independent. This is closely related to the *right endpoint* of the underlying distribution: DEF 3.3

$$x_F := \sup\{x \in \mathbb{R} : F(x) < 1\}$$

Heuristically, for large n , we expect the worst-case loss to capture the largest attainable value under F . In particular, one can show that $M_n \rightarrow x_F$ in probability. DEF 3.4

We say that X with distribution F belongs to the *maximum domain of attraction* of a random, non-degenerate distribution H if there exist sequences (a_n) and $(b_n) > 0$ with DEF 3.5

$$F^n(b_n x + a_n) \rightarrow H(x)$$

for all continuity points x of H . In this event, we write $X \in \text{MDA}(H)$ or $F \in \text{MDA}(H)$.

For independent samples, $X \in \text{MDA}(H)$ if and only if $\frac{M_n - a_n}{b_n} \rightsquigarrow H$. PROP 3.1

F^n is the distribution of M_n . Letting H be distributed by H , we rearrange: PROOF.

$$\mathbb{P}(M_n \leq b_n x + a_n) = \mathbb{P}\left(\frac{M_n - a_n}{b_n} \leq x\right) \rightarrow \mathbb{P}(H \leq x)$$

Note that $b_n > 0$ ensures that the division step preserves the inequality. □

Eg. 3.1 Let F be uniform. Then $F(x) = x$ and $F^n(x) = x^n$. Then, for $x \in \mathbb{R}$

$$\mathbb{P}(n(M_n - 1) \leq x) = \mathbb{P}\left(M_n \leq \frac{x + n}{n}\right) = \left(\frac{x + n}{n}\right)^n = \left(\frac{x}{n} + 1\right)^n$$

With a change of variables $nx = m$, we recognize that this converges to e^x . However, when $x \geq 0$, we see that $M_n - 1 \leq 0$, so it must be that $\mathbb{P}(n(M_n - 1) \leq$

$x) = 1$. We conclude that

$$\mathbb{P}(n(M_n - 1) \leq x) \rightarrow \begin{cases} e^x & x < 0 \\ 1 & x \geq 0 \end{cases} =: H$$

so $F \in \text{MDA}(H)$

Eg. 3.2 Let F be $\text{Exp}(1)$, i.e. $F(x) = 0$ for $x < 0$ and $1 - e^{-x}$ for $x \geq 0$. Then

$$\mathbb{P}(M_n - \ln(n) \leq x) \rightarrow e^{-e^{-x}}$$

We follow a similar argument, recovering the limit definition of e . We call this limit the *Gumbel distribution*, which we denote Λ . In other words $F \in \text{MDA}(\Lambda)$.

Eg. 3.3 When F is $\text{Beta}(1, \alpha)$ on $(0, 1)$, we have

$$\mathbb{P}(n^{1/\alpha}(M_n - 1) \leq x) \rightarrow \begin{cases} e^{-(-x)^\alpha} & x < 0 \\ 1 & x \geq 0 \end{cases}$$

which we call the *Weibull distribution* and denote Ψ_α .

Eg. 3.4 When F is $\text{Pareto}(1, \alpha)$ on $\mathbb{R}$,

$$\mathbb{P}\left(\frac{M_n + 1}{n^{1/\alpha}} \leq x\right) \rightarrow \begin{cases} 0 & x < 0 \\ e^{-\frac{1}{x^\alpha}} & x \geq 0 \end{cases}$$

which we call the *Fréchet distribution*, which we denote by Φ_α .

We say that variables are of the same *type* if their distributions are equivalent up to translation and scaling, i.e. if $H^*(x) = H\left(\frac{x-a}{b}\right)$, then $H^* \sim H$.

PROP 3.2 Suppose the sequences (a_n) and $(b_n) > 0$ satisfy $\frac{M_n - a_n}{b_n} \rightsquigarrow H$. Let (a_n^*) and $(b_n^*) > 0$ satisfy

$$\lim_{n \rightarrow \infty} \frac{b_n}{b_n^*} = b \in \mathbb{R} \quad \lim_{n \rightarrow \infty} \frac{a_n - a_n^*}{b_n^*} = a \in \mathbb{R}$$

Then $\frac{M_n - a_n^*}{b_n^*} \rightsquigarrow H^*$, where $H^*(x) = H\left(\frac{x-a}{b}\right)$. The converse also holds.

PROOF.

We'll prove ($\Rightarrow$). It comes down to an unwrapping of limits. In particular, writing $\lim_{n \rightarrow \infty} a_n^* = a_n - ab_n^*$, we get

$$\begin{aligned} \mathbb{P}\left(\frac{M_n - a_n^*}{b_n^*} \leq y\right) &\rightarrow \mathbb{P}\left(b \frac{M_n - a_n^*}{b_n} \leq y\right) \rightarrow \mathbb{P}\left(b \frac{M_n - a_n + b_n^* a}{b_n} \leq y\right) \\ &= \mathbb{P}\left(\frac{M_n - a_n}{b_n} \leq \frac{y-a}{b}\right) \rightarrow H\left(\frac{y-a}{b}\right) = H^*(y) \quad \square \end{aligned}$$

In other words, if $F \in \text{MDA}(H)$ and $F \in \text{MDA}(H^*)$, then H and H^* are equivalent up to scaling.

Given a *shape parameter* $\xi \in \mathbb{R}$, the following distribution

DEF 3.10

$$H_\xi(x) = \begin{cases} \exp(1 - (1 + \xi x)^{-1/\xi}) & \xi \neq 0 \\ \exp(-\exp(-x)) & \xi = 0 \\ 0 & \text{otherwise} \end{cases} \quad x \in \mathbb{R} : 1 + \xi x > 0$$

Note that the limit of $H_\xi(x)$ as $\xi \rightarrow 0$ is consistent with the $\xi = 0$ case

is called the *extreme value distribution*. Furthermore, for mean $\mu \in \mathbb{R}$ and standard deviation $\sigma \in \mathbb{R}_{>0}$, we define

DEF 3.11

$$H_{\xi, \mu, \sigma}(x) = H_\xi\left(\frac{x - \mu}{\sigma}\right) : x \in \mathbb{R}$$

which we call the *generalized extreme value distribution*, or GEV.

DEF 3.12

Importantly, the GEV specializes to the following cases:

$$H_\xi = \begin{cases} \text{Fréchet} & \xi > 0 \\ \text{Gumbel} & \xi = 0 \\ \text{Weibull} & \xi < 0 \end{cases}$$

Typically, Fréchet distributions are heavy tailed (e.g. student-t or Pareto) while Weibull distributions are light tailed (e.g. Beta). Gumbell distributions are somewhere in between, and encapsulate Normal, Log-Normal, Chi-Squared, Exponential, and Gamma distributions.

If $F \in \text{MDA}(H_\xi)$, then $x_F = \infty$ if $\xi > 0$ and $x_F < \infty$ if $\xi < 0$.

PROP 3.3

Theorem 3.1 Fisher-Teppet

If $F \in \text{MDA}(H)$ for some non-degenerate distribution H , then H must be GEV distributed, and we write $H = H_{\xi, \mu, \sigma}$. In particular, μ, σ are dependent on the choice of normalizing sequences $(a_n), (b_n)$, while ξ is unique to F , as established in [Prop 3.2](#). One can show that we can choose normalizing sequences such that $H = H_\xi$.

We are curious about when $F \in \text{MDA}(H_\xi)$.

Theorem 3.2 Gnedenko's Theorem

$F \in \text{MDA}(H_\xi)$ with $\xi > 0$ if and only if $\bar{F}$ is *regularly varying* with index $-\frac{1}{\xi}$, i.e.

DEF 3.13

$$\lim_{x \rightarrow \infty} \frac{\bar{F}(xt)}{\bar{F}(x)} = t^{-\frac{1}{\xi}} \quad \forall t > 0$$

Equivalently,

$$\bar{F}(x) = x^{-\frac{1}{\xi}} L(x) \quad \text{where} \quad \lim_{x \rightarrow \infty} \frac{L(tx)}{L(x)} = 1 \quad \forall t > 0$$

If $F \in \text{MDA}(H_\xi)$ with $\xi > 0$, then $\mathbb{E}[X^k] = \infty$ for all $k > \frac{1}{\xi}$.

PROP 3.4

Theorem 3.3 von Mises Criterion (Fréchet)

Suppose F has density f . Then if

$$\lim_{x \rightarrow \infty} \frac{xf(x)}{\bar{F}(x)} = \frac{1}{\xi}$$

$F \in \text{MDA}(H_\xi)$ with $\xi > 0$.

Theorem 3.4 von Mises Criterion (Gumbel)

Let F be twice differentiable on some interval (z, x_F) . Assume also, on this interval, that $F' = f > 0$ and $F'' < 0$, i.e. F is concave on (z, x_F) . If

$$\lim_{x \rightarrow x_F} \frac{\bar{F}(x)F''(x)}{f^2(x)} = -1$$

then F is in the Gumbel domain of attraction.

PROP 3.5 If $F \in \text{MDA}(H_\xi)$ with $\xi < 0$, then $\mathbb{E}[X^k] < \infty$ for all $k > 0$.

BLOCK MAXIMA METHOD

Suppose $F \in \text{MDA}(H_\xi)$, and let $X_1, \dots, X_N$ be samples from F . We wish to estimate the parameter ξ . The following steps provide a method for doing so, collectively called the

DEF 3.14 *block maxima method*:

1. Choose m, n such that $nm = N$. We call n the "block size."
2. Separate the data samples into "blocks" as follows

$$\underbrace{X_1, \dots, X_n}_{M_{n1}}, \underbrace{X_{n+1}, \dots, X_{2n}}_{M_{n2}}, \dots, \underbrace{X_{(m-1)n+1}, \dots, X_{mn}}_{M_{nm}}$$

3. If n is large enough, $M_{ni} : i \in [m]$ are IID, with $M_{ni} \approx H_{\xi, \mu, \sigma}$. One can estimate these parameters via MLE.

When n becomes large, we decrease bias (i.e. M_{ni} is more representative of an asymptotic maxima) but increase variance (since we yield less blocks). Conversely, when m becomes large, we increase bias and decrease variance.

Return Level and Period

DEF 3.15 The *return level* of k blocks is the value r that is exceed by, on average, by one out of k blocks of size n .

For instance, if we form rainfall data into year-long block intervals, we might ask for the return value of 100 years, indicating the expected rainfall that we would expect to see once out of every 100 years.

If blocks $M_{in} \stackrel{iid}{\sim} H_{\xi, \mu, \sigma}$, then the k -block return level is

PROP 3.6

$$r = H_{\xi, \mu, \sigma}^{-1} \left(1 - \frac{1}{k} \right) = \begin{cases} \mu + \frac{\sigma}{\xi} \left(e^{-\xi \ln(-\ln(1 - \frac{1}{k}))} - 1 \right) & \xi \neq 0 \\ \mu + \sigma \left(-\ln(-\ln(1 - \frac{1}{k})) \right) & \xi = 0 \end{cases}$$

Fix some arbitrary r . Then, consider the index Z of the first trial in which a block M_{in} exceeds r . With $\mathbb{P}(r < M_{in}) = 1 - H(r) =: p$, this is geometric with probability p , i.e.

PROOF.

$$\mathbb{P}(Z = k) = (1 - p)^{k-1} p$$

Then, the expected number of trials needed at level r is $\mathbb{E}[Z] = \frac{1}{p} = \frac{1}{1-H(r)}$. Hence, the return value r for k blocks satisfies $k = \frac{1}{1-H(r)} \iff r = H^{-1} \left(1 - \frac{1}{k} \right)$, as desired. $\square$

Similarly, the *return period* of some value r is the average number k of blocks of size n required to exceed r .

DEF 3.16

The return period of r is $\frac{1}{1-H_{\xi, \mu, \sigma}(r)}$.

PROP 3.7

This follows directly from the proof of [Prop 3.6](#). $\square$

PROOF.

EXCESS DISTRIBUTION

Let X follow distribution F . Let x_F be the right endpoint of F . For $u < x_F$, called the *threshold*, we define

DEF 3.17

$$F_u(y) := \mathbb{P}(X - u \leq y | X > u) : y \geq 0$$

called the *excess distribution* of X with respect to u . We call

DEF 3.18

$$e(u) := \mathbb{E}[X - u | X > u]$$

the *mean excess function* of X with respect to u .

DEF 3.19

Define

$$G_{\xi}(x) = \begin{cases} 1 - (1 + \xi x)^{-\frac{1}{\xi}} & \xi \neq 0 \\ 1 - e^{-x} & \xi = 0 \end{cases} \quad G_{\beta, \xi}(x) = G_{\xi} \left(\frac{x}{\beta} \right)$$

which is non-zero for $x \geq 0$, $\xi \geq 0$, or for $x \in [0, -\frac{1}{\xi}]$, $\xi < 0$, and zero elsewhere. We call $G_{\beta, \xi}$ the *generalized Pareto distribution*, or GPD.

DEF 3.20

Note that $G_{\beta, \xi}$ reduces to $\text{Exp}(\beta)$ when $\xi = 0$. Also note that

$$\mathbb{E}[G_{\beta, \xi}] = \begin{cases} \frac{\beta}{1-\xi} & \xi < 1 \\ \infty & \xi \geq 1 \end{cases}$$

Thus, if $F_u \approx G_{\beta, \xi}$, then $e(u) \approx \mathbb{E}[G_{\beta, \xi}]$.

Theorem 3.5 Pickands-Balkemade-de Haan Theorem

For $\xi \in \mathbb{R}$, the following are equivalent:

1. $F \in \text{MDA}(H_\xi)$
2. There exists some measurable function $\beta : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\lim_{u \rightarrow x_F} \sup_{x \in [0, x_F - u)} |F_u(x) - G_{\beta(u), \xi}(x)| = 0$$

In short, and in practice, if $u \approx x_F$, then $F_u \approx G_{\beta, \xi}(x)$, for some β , when $F \in \text{MDA}(H_\xi)$.

PROP 3.8 If $X \sim G_{\beta, \xi}$, then the excess distribution of X is also GPD. In particular, $F_u = G_{\beta + \xi u, \xi}$. It follows that, for $\xi < 1$, $e(u) = \frac{\beta + \xi u}{1 - \xi}$.

Risk Estimation using GPD

PROP 3.9 When u is high enough (i.e. $u \approx x_F$) and $F \in \text{MDA}(H_\xi)$, $F_u \approx G_{\beta, \xi}$. Then, for $x \geq u$,

$$\bar{F}(x) = \bar{F}(u) \bar{G}_{\beta, \xi}(x - u)$$

PROOF.

We unwrap as follows:

$$\begin{aligned} \bar{F}(x) &= \mathbb{P}(X > x) = \mathbb{P}(X > x | X > u) \mathbb{P}(X > u) \\ &= \bar{F}(u) \mathbb{P}(X - u > x - u | X > u) = \bar{F}(u) \bar{F}_u(x - u) \\ &\approx \bar{F}(u) \bar{G}_{\beta, \xi}(x - u) \quad \square \end{aligned}$$

As a consequence, we can extrapolate into the tail of X using u as an anchor. We'll apply this to the original cause of VaR_α and ES_α estimation.

PROP 3.10 Let $F \in \text{MDA}(H_\xi)$ and $u : F_u \approx G_{\beta, \xi}$. Let $\alpha \geq F(u)$. Then

$$\text{VaR}_\alpha(X) = u + \begin{cases} \frac{\beta}{\xi} \left[\left(\frac{1-\alpha}{\bar{F}(u)} \right)^{-\xi} - 1 \right] & \xi \neq 0 \\ -\beta \ln \left(\frac{1-\alpha}{\bar{F}(u)} \right) & \xi = 0 \end{cases}$$

PROOF.

Since $\alpha \geq F(u)$, $F^{-1}(\alpha) = \text{VaR}_\alpha(X) \geq u$. Then, by [Prop 3.9](#),

$$\bar{F}(\text{VaR}_\alpha(X)) = \bar{F}(u) \bar{G}_{\beta, \xi}(\text{VaR}_\alpha(X) - u)$$

We have $\bar{F}(\text{VaR}_\alpha(X)) = 1 - \mathbb{P}(X \leq \text{VaR}_\alpha(X)) = 1 - \alpha$, so

$$1 - \alpha = \bar{F}(u) \bar{G}_{\beta, \xi}(\text{VaR}_\alpha(X) - u)$$

Rearranging, we find

$$\frac{1 - \alpha}{\bar{F}(u)} = \bar{G}_{\beta, \xi}(\text{VaR}_\alpha(X) - u) \implies \text{VaR}_\alpha(X) = u + \bar{G}_{\beta, \xi}^{-1}\left(\frac{1 - \alpha}{\bar{F}(u)}\right)$$

Writing $\bar{G}^{-1}$ explicitly gives the result. $\square$

Let $F \in \text{MDA}(H_\xi)$ and $u : F_u \approx G_{\beta, \xi}$. Let $\alpha \geq F(u)$. Then

PROP 3.11

$$\text{ES}_\alpha(X) = \begin{cases} \frac{1}{1-\xi} \text{VaR}_\alpha(X) + \frac{\beta - \xi u}{1-\xi} & \xi < 1 \\ \infty & \xi \geq 1 \end{cases}$$

When $F \in \text{MDA}(H_\xi)$ and $u : F_u \approx G_{\beta, \xi}$, the shortfall-to-quantile ratio is

PROP 3.12

$$\lim_{\alpha \rightarrow 1} \frac{\text{ES}_\alpha(X)}{\text{VaR}_\alpha(X)} = \frac{1}{1 - \xi}$$

for $\xi \in (0, 1]$ (and ∞ otherwise). Note that, when $\xi = 0$, we also require that $x_F = \infty$. This is guaranteed when $\xi > 0$, but not when $\xi = 0$. In either case, the condition $x_F = \infty$, so that $\text{VaR}_\alpha(X) \rightarrow \infty$, is necessary for the limit to converge as stated.

Peaks-Over-Threshold Method

When choosing a threshold u , we would like $F_u \approx G_{\beta, \xi}$, per [Thm 3.5](#). But we also would like u to be as small as possible, so that empirical estimations of $\bar{F}(u)$ are as accurate as possible, per [Prop 3.9](#) and its following uses in estimating VaR and ES.

Given data $X_1, \dots, X_n$:

1. For any $u > 0$, find $N_u := \{X_i : X_i > u\}$
2. Relabel $N_u = \{\tilde{X}_1, \dots, \tilde{X}_N\}$
3. Calculate the exceedences $Y_i := \tilde{X}_i - u$ for each $i \in [N]$
4. Fit $G_{\beta, \xi}$ to the data $\{Y_1, \dots, Y_N\}$.

We then use the approximation $\bar{F}(x) = \bar{F}(u) \bar{G}_{\beta, \xi}(x - u)$ to compute VaR and ES, per the above discussion. With empirical data, we can estimate $\bar{F}(u) \approx |N_u|/n$. Altogether, is called the *peaks-over-threshold method*.

DEF 3.21

If $F_u = G_{\beta, \xi}$, then, for $v \geq u$, $F_v = G_{\beta + \xi(v-u), \xi}$.

PROP 3.13

To choose u , then, we have

$$e(v) = \mathbb{E}[X - v | X > v] = \mathbb{E}[F_v] \approx \mathbb{E}[G_{\beta + \xi(v-u), \xi}] = \frac{\beta + \xi(v - u)}{1 - \xi}$$

which is positively linear in u . Thus, we pick the smallest possible u such that $e(v)$ is positively linear in v for $v > u$. For simplicity, this is best completed graphically.

IV Backtesting

STATISTICAL TESTS FOR VAR AND ES

Suppose we have generated estimates for VaR_α^t and ES_α^t using time series models, with or without extreme value theory-based methods. Recall that, empirically, we estimate

$$\text{VaR}_\alpha^t = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \text{VaR}_\alpha(Z_{t+1}) \quad \text{and} \quad \text{ES}_\alpha^t = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \text{ES}_\alpha(Z_{t+1})$$

DEF 4.1 If we only utilize data $X_s : s \leq t$, then $\hat{\mu}_{t+1}$ and $\hat{\sigma}_{t+1}$, and the associated risk measure are called *out-of-sample estimates*. Backtesting aims to quantify the performance of out-of-samples risk measure estimates.

VaR

In our context, we let $L_1, \dots, L_t$ be losses, and write

$$\text{VaR}_\alpha^t = F_{L_{t+1}|\mathcal{F}_t}^{-1}(\alpha)$$

DEF 4.2 Let $\ell_{t+1} = \mathbb{1}_{L_{t+1} > \text{VaR}_\alpha^t}$ be *violation indicators*. Then $\mathbb{E}[\ell_{t+1}|\mathcal{F}_t] = 1 - \alpha$ so $\ell_{t+1} \sim \text{Ber}(1 - \alpha)$. One can show that any sequence of binary variables whose conditional distribution is Bernoulli is IID. Thus:

PROP 4.1 $\ell_{t+1} \stackrel{iid}{\sim} \text{Ber}(1 - \alpha)$. We conclude that $\sum_{t=1}^m \ell_{t+1} \sim \text{Bin}(m, 1 - \alpha)$, and, if $\ell_{t+1} > \text{VaR}_\alpha^t$ occurs at times $\{T_1, \dots, T_m\}$, the interarrival times $T_i - T_{i-1} \sim \text{Geom}(1 - \alpha)$ for $i \in [m]$.

PROP 4.2 Suppose we have provided empirical estimates $\widehat{\text{VaR}}_\alpha^t : t \in [m]$, and thus empirical violation indicators $\hat{\ell}_{t+1} : t \in [m]$. Then, under the null $H_0 : \widehat{\text{VaR}}_\alpha^t = \text{VaR}_\alpha^t$, in which case $\hat{\ell}_{t+1}$ are Bernoulli, we have

$$Z_m = \frac{\sum_{t=1}^m \hat{\ell}_{t+1} - m(1 - \alpha)}{\sqrt{m\alpha(1 - \alpha)}} \sim \mathcal{N}(0, 1)$$

DEF 4.3 This is called the *Kupiec statistic*. Equivalently, we can use

$$L_m = -2 \log \left(\frac{\alpha^{m - \sum_{t=1}^m \hat{\ell}_{t+1}} (1 - \alpha)^{\sum_{t=1}^m \hat{\ell}_{t+1}}}{\hat{\alpha}^{m - \sum_{t=1}^m \hat{\ell}_{t+1}} (1 - \hat{\alpha})^{\sum_{t=1}^m \hat{\ell}_{t+1}}} \right) \sim \chi_1^2 \quad \text{where} \quad \hat{\alpha} = 1 - \frac{\sum_{t=1}^m \hat{\ell}_{t+1}}{m}$$

When $\alpha \approx 1$ and m is large, ℓ_{t+1} are Bernoulli with many trials and small probability. Thus, these become events in a Poisson process with rate $(1 - \alpha)\Delta t$, and so their interarrival times, i.e. $T_i - T_{i-1}$, are $\text{Exp}((1 - \alpha)\Delta t)$ distributed.

DEF 4.4 Following this discussion, under the same null $H_0 : [\hat{\ell}_{t+1} \text{ are Bernoulli}]$, the *Christoffersen-Pelletier statistic* tests for exponential interarrival times. We do not cover the statistic itself.

In the discrete case, we said these were $\text{Geom}(1 - \alpha)$ distributed.

Heuristically, while the Kupiec test ensures that we get the correct violation rate, the Christoffersen-Pelletier test ensures that violations do not occur in clusters.

ES

Given losses $L_1, \dots, L_t$, suppose we can write $L_t = \mu_t + \sigma_t Z_t$, where $Z_t \sim \text{SWN}(0, 1)$ and μ_t, σ_t are measurable w.r.t. $\mathcal{F}_t$. Then consider

$$\text{ES}_\alpha^t = \mathbb{E}[L_{t+1} | L_{t+1} > \text{VaR}_\alpha^t, \mathcal{F}_t]$$

Since $L_t = \mu_t + \sigma_t Z_t$, we find that $\text{ES}_\alpha^t = \mu_{t+1} + \sigma_{t+1} \text{ES}_\alpha(Z_{t+1})$.

Similar to the violation indicators ℓ_{t+1} , consider the *violation residuals*:

DEF 4.5

$$K_{t+1} = \left(\frac{L_{t+1} - \text{ES}_\alpha^t}{\text{ES}_\alpha^t - \mu_{t+1}} \right) \mathbb{1}_{L_{t+1} > \text{VaR}_\alpha^t}$$

where $\mu_{t+1} = \mathbb{E}[L_{t+1} | \mathcal{F}_t]$ is the conditional mean.

When $L_t = \mu_t + \sigma_t Z_t$,

PROP 4.3

$$K_{t+1} = \frac{Z_{t+1} - \text{ES}_\alpha(Z_{t+1})}{\text{ES}_\alpha(Z_{t+1})} \mathbb{1}_{Z_{t+1} > \text{VaR}_\alpha(Z_{t+1})}$$

and thus, if the distribution Z is continuous, $\mathbb{E}[K_{t+1} | \mathcal{F}_t] = 0$.

Suppose we generate empirical expected shortfalls, $\widehat{\text{ES}}_\alpha^t : t \in [m]$, and thus empirical violation residuals $\widehat{K}_{t+1} : t \in [m]$. Under the null $H_0 : \widehat{\text{ES}}_\alpha^t = \text{ES}_\alpha^t$, it should be that $\frac{\sum_{t=1}^m \widehat{K}_{t+1}}{m} = 0$. One can check this using, for instance, a one-sided t -test.

ELICITABILITY

Suppose $\rho : \mathcal{M} \rightarrow \mathbb{R}$ is a functional on loss distributions $L \mapsto \rho(L)$. Typically, ρ is a risk measure (recall [Def 1.23](#)), but not necessarily so.

Given data $X_1, \dots, X_t$, we wish to estimate $\rho_t := \rho(L_{t+1} | \mathcal{F}_t)$. Suppose we have generated estimates $\hat{\rho}_t^{(1)}, \dots, \hat{\rho}_t^{(j)}$ via a collection of forecasting strategies. Our goal is to choose the best forecasting method. For this, we turn to the theory of elicibility.

Elicitable Functions

Let $S : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_{>0}$. We call S a *scoring function* if, for any $y, \ell \in \mathbb{R}$, we have

DEF 4.6

1. $S(y, \ell) \geq 0$ and $S(y, \ell) = 0 \iff y = \ell$
2. When ℓ is fixed, $S(y, \ell)$ is decreasing when $y < \ell$ and increasing when $y > \ell$.
3. When ℓ is fixed, S is continuous in y .

For instance, $S(y, \ell) = (y - \ell)^2$ is a scoring function. Scoring functions are *not quite* metrics, but they are close.

Let $\rho : \mathcal{M} \rightarrow \mathbb{R}$ be a loss functional. It is called *elicitable* if there exists some scoring function S such that, for any $L \in \mathcal{M}$,

DEF 4.7

We are missing the
triangle inequality
and symmetry

1. $\mathbb{E}[S(y, L)] < \infty \forall y \in \mathbb{R}$
2. $\rho(L)$ minimizes the expected score, i.e.

$$\rho(L) = \operatorname{argmin}_{y \in \mathbb{R}} \mathbb{E}[S(y, L)]$$

In this case, we say that ρ is strictly consistent for the scoring function S .

Eg. 4.1 The mean $\rho(L) = \mathbb{E}[L]$ is elicitable on the space of variables with finite variance, with the scoring function $(y - \ell)^2$.

Eg. 4.2 The median $\rho(L) = F_L^{-1}(\frac{1}{2})$ is elicitable on the space of variables such that $\mathbb{E}[|L|] < \infty$ and F_L is strictly increasing. The scoring function is $|y - \ell|$.

Eg. 4.3 As a generalization, for $\alpha \geq \frac{1}{2}$, $\operatorname{VaR}_\alpha(L)$ is elicitable on the space of variables such that $\mathbb{E}[|L|] < \infty$ and F_L is strictly increasing. The scoring function is

$$S(y, \ell) = |\mathbb{1}_{\ell \leq y} - \alpha| \cdot |\ell - y| = \begin{cases} \alpha(\ell - y) & y \leq \ell \\ (1 - \alpha)(y - \ell) & y > \ell \end{cases}$$

Fix an elicitable function $\rho(L)$ with scoring function S and estimates $\hat{\rho}_t^{(1)}, \dots, \hat{\rho}_t^{(J)}$ for $\rho(L_{t+1}|\mathcal{F}_t)$. Then, since

$$\rho(L) = \operatorname{argmin}_{y \in \mathbb{R}} \mathbb{E}[S(y, L)]$$

it would be principled to choose a strategy $j \in [J]$ such that $\mathbb{E}[S(\hat{\rho}_t^{(j)}, L_{t+1})]$ is minimized. To compute this expectation, we use strategy j to compute estimates for all available data, i.e. find

$$\operatorname{argmin}_{j \in [J]} \left(\frac{1}{m} \sum_{t \in [m]} S(\hat{\rho}_t^{(j)}, L_{t+1}) \right)$$

Eg. 4.4 Recall [Def 2.38](#). This was the estimate $\hat{\mu}_{t+h}$ which minimized $\mathbb{E}[(X_{t+1} - P)^2]$ over all $\mathcal{F}_t$ -measurable variables P . We conclude that the conditional mean $\hat{\mu}_{t+h} = \mathbb{E}[X_{t+h}|\mathcal{F}_t]$ is an elicitable function with survivor function $(y - \ell)^2$.

PROP 4.4 ES_α is not elicitable. One can show this by computing e_α , as defined below.

PROP 4.5 Let $\rho(L)$ be monotonic and translation-invariant. Then $\rho(L)$ is elicitable and coherent if and only if it is consistent for the scoring function

$$S(y, \ell) = |\mathbb{1}_{\ell \leq y} - \alpha|(\ell - y)^2$$

for some $\alpha \in [\frac{1}{2}, 1]$.

This proposition suggests the following mysterious, coherent, elicitable risk measure:

$$e_\alpha = \operatorname{argmin}_{y \in \mathbb{R}} \mathbb{E}[|\mathbb{1}_{\ell \leq y} - \alpha|(\ell - y)^2]$$

DEF 4.8 called the *expectile*.

For $\alpha = 0.5$,
 $e_\alpha = \mathbb{E}[L]$.

We conclude that neither VaR (since it is not coherent) nor ES is e_α , i.e. the unique coherent and elicitable risk measure. So why don't we use it? Because solutions to e_α are not always tractable.

V Multivariate Modeling

For much of this course, we have considered log returns to be univariate risk factor changes. Recall, however, that we allow ourselves to consider multivariate risk factor changes, e.g. $\mathbf{X}_1, \dots, \mathbf{X}_t$. For instance, these may be joint log returns,

$$\mathbf{X}_t = \left\langle \log \left(\frac{S_{t+1}^{(A)}}{S_t^{(A)}} \right), \log \left(\frac{S_{t+1}^{(B)}}{S_t^{(B)}} \right) \right\rangle$$

We can develop distributional (unconditional) or dynamic (conditional) models. In the former, we calibrate some suitable stationary multivariate model, and, in the latter, we develop a multivariate time series model in much the same spirit as ARMA-GARCH models. For our purposes, we will only consider a distributional approach.

MULTIVARIATE STATISTICS REFRESHER

Distributions and Densities

Suppose $\mathbf{X} = \langle X_1, \dots, X_n \rangle$ is a vector of variables. We remind ourselves of the following:

- $F_{\mathbf{X}}(x_1, \dots, x_n) = \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n)$ is the joint CDF.
- $\bar{F}_{\mathbf{X}}(x_1, \dots, x_n) = \mathbb{P}(X_1 > x_1, \dots, X_n > x_n)$ is the joint survivor function. Note that, in the multivariate case, $\bar{F}_{\mathbf{X}} \neq 1 - F_{\mathbf{X}}$, necessarily.
- $F_{X_i}(x) = F_i(x) = \mathbb{P}(X_i \leq x)$ is the marginal CDF at index i . We can recover it from the joint CDF via

$$F_{X_i}(x) = F_{\mathbf{X}}(\infty, \dots, \underbrace{x}_{i\text{-th index}}, \dots, \infty)$$

Note that, in truth, these are coordinate-wise limits to infinity.

- Analogously, if $\mathbf{X} = \langle \mathbf{A}, \mathbf{B} \rangle$, (coordinate-wise), then a higher-order marginal CDF of $\mathbf{A}$, equivalent to the joint CDF over variables in $\mathbf{A}$, can be generated by "setting" all arguments in F corresponding to variables in $\mathbf{B}$ to ∞ , as above.
- A joint CDF $F_{\mathbf{X}}$ is called absolutely continuous if

$$F_{\mathbf{X}}(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(x_1, \dots, x_n) dx_n \cdots dx_1$$

for some non-negative function f , which, in this case, is called the joint density. The joint density need not exist, but we will avoid such distributions here.

- If F_X is absolutely continuous, then we can recover the marginal densities as follows

$$f_i(x) = \int_{\mathbb{R}} \cdots \int_{\mathbb{R}} f(x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n) dx_n \cdots dx_1$$

which, to be sure, is a $n - 1$ dimensional integral. When $\mathbf{X} = \langle \mathbf{A}, \mathbf{B} \rangle$, we can similarly form higher-order marginal densities by "integrating out" coordinates not used. Note that, if $X_1, \dots, X_n$ are given densities $f_1, \dots, f_n$, it may not be the case that $\mathbf{X}$, at the end of the day, is absolutely continuous.

- Suppose $\mathbf{X} = \langle \mathbf{A}, \mathbf{B} \rangle$ is absolutely continuous. Then the conditional density of $\mathbf{B}$ with respect to $\mathbf{A}$ is given by

$$f_{\mathbf{B}|\mathbf{A}}(\mathbf{b}|\mathbf{a}) = \frac{f_X(\langle \mathbf{a}, \mathbf{b} \rangle)}{f_A(\mathbf{a})}$$

Independence

- Let $\mathbf{X} = \langle \mathbf{A}, \mathbf{B} \rangle$. Then we say that $\mathbf{A}$ and $\mathbf{B}$ are independent, and write $\mathbf{A} \perp \mathbf{B}$, if $F_X(\langle \mathbf{a}, \mathbf{b} \rangle) = F_A(\mathbf{a})F_B(\mathbf{b})$. This holds equivalently if the joint density splits, $f_X(\langle \mathbf{a}, \mathbf{b} \rangle) = f_A(\mathbf{a})f_B(\mathbf{b})$.
- Similarly, $X_1, \dots, X_n$ are mutually independent, and write $X_i \perp X_j : i \neq j$, if

$$F_X(x_1, \dots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n)$$

This holds equivalently if the joint density splits, i.e. $f_X(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$.

Moments

- Given some multivariate $\mathbf{X}$, we write

$$\mu := \mathbb{E}[\mathbf{X}] = \langle \mathbb{E}[X_1], \dots, \mathbb{E}[X_n] \rangle$$

- Similarly, we define the covariance matrix

$$\Sigma := \text{cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T] = \mathbb{E}[\mathbf{X}\mathbf{X}^T] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^T$$

which satisfies $\Sigma_{i,j} = \text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])] = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j]$.

PROP 5.1

- Note that the covariance matrix Σ is always positive semidefinite (i.e. $x^T \Sigma x \geq 0$ and $\Sigma = \Sigma^T$). If it is positive definite (i.e. $x^T \Sigma x > 0$), then there exists some unique lower-triangular matrix $\Sigma^{\frac{1}{2}}$, called the *Cholesky factor*, such that $\Sigma^{\frac{1}{2}} \Sigma^{\frac{1}{2}T} = \Sigma$.

DEF 5.1

- Similarly, the correlation matrix is

$$P_{i,j} = \frac{\text{Cov}(X_i, X_j)}{\sqrt{\text{Var}(X_i)\text{Var}(X_j)}}$$

Note that, if Δ is the matrix with 0s everywhere except $\Delta_{i,i} = \sqrt{\Sigma_{i,i}}$, then we can write

$$P = \Delta^{-1} \Sigma \Delta^{-1} =: \mathcal{R}(\Sigma)$$

- Given $\mathbf{X}_1, \dots, \mathbf{X}_k$, we can generate the usual sample estimates

$$\bar{\mathbf{X}} = \frac{1}{k} \sum_{i \in [k]} \mathbf{X}_i \quad \text{and} \quad (S)_{ij} = \frac{1}{k} \sum_{i \in [k]} (\mathbf{X}_i - \bar{\mathbf{X}})^T (\mathbf{X}_i - \bar{\mathbf{X}})$$

for $\boldsymbol{\mu}$ and Σ , respectively.

- $\mathbb{E}[\mathbf{b} + A\mathbf{X}] = \mathbf{b} + A\mathbb{E}[\mathbf{X}]$ and $\text{Cov}(\mathbf{b} + A\mathbf{X}) = A\text{Cov}(\mathbf{X})A^T$.

MULTIVARIATE NORMAL

We envisage the multivariate normal in the following stochastic (non-canonical) way. We say that $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$ is *multivariate normal* if there exists some $A \in M_{d \times k}(\mathbb{R})$ satisfying $AA^T = \Sigma$, with

DEF 5.2

$$\mathbf{X} \stackrel{d}{=} \boldsymbol{\mu} + A\mathbf{Z}$$

where $\mathbf{Z} = \langle Z_1, \dots, Z_d \rangle$ is a row of IID standard normal variables, i.e. $Z_i \stackrel{iid}{\sim} \mathcal{N}(0, 1)$.

$\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{X}) = \Sigma$.

PROP 5.2

$$\text{Cov}(\mathbf{X}) = A\text{Cov}(\mathbf{Z})A^T = AIA^T = AA^T = \Sigma.$$

□

PROOF.

If $d \leq k$, we can construct $\mathbf{X}$ with exactly $k = d$ standard normals. In this case, Σ is invertible and positive definite, so we can write $A = \Sigma^{\frac{1}{2}}$. From now on, all of our multivariate normal covariance matrices will be positive definite and invertible.

PROP 5.3

To compute a multivariate normal $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$, then, we do the following:

1. Compute the Cholesky factor $\Sigma^{\frac{1}{2}}\Sigma^{\frac{1}{2}T} = \Sigma$.
2. Simulate IID standard normals, $Z_1, \dots, Z_n$, and set $\mathbf{Z} = \langle Z_1, \dots, Z_n \rangle$.
3. Set $\mathbf{X} = \boldsymbol{\mu} + \Sigma^{\frac{1}{2}}\mathbf{Z}$

The multivariate *characteristic function* $\phi_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}[e^{i\mathbf{t}^T \mathbf{X}}]$, where $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$, is

DEF 5.3

$$\phi_{\mathbf{X}}(\mathbf{t}) = \exp(i\mathbf{t}^T \boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^T \Sigma \mathbf{t})$$

The proof of this follows from the computation $\phi_{\mathbf{Z}}(\mathbf{t}) = \exp(-\frac{1}{2}\mathbf{t}^T \mathbf{t}) := \psi(\mathbf{t})$.

Let $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$, and let $\mathbf{X} = \langle \mathbf{X}_1, \mathbf{X}_2 \rangle$, where $|\mathbf{X}_1| = k$. Writing

PROP 5.4

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix} \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

along the obvious boundaries dictated by k and $d - k$, we have that

$$\mathbf{X}_1 \sim \mathcal{N}_k(\boldsymbol{\mu}_1, \Sigma_{11}) \quad \text{and} \quad \mathbf{X}_2 \sim \mathcal{N}_{d-k}(\boldsymbol{\mu}_2, \Sigma_{22})$$

PROP 5.5 If $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$ and $B \in M_{k \times d}$, then $B\mathbf{X} + \mathbf{b} \sim \mathcal{N}_k(B\boldsymbol{\mu} + \mathbf{b}, B\Sigma B^T)$.

PROP 5.6 $\mathbf{X}$ is some multivariate normal if and only if $\mathbf{a}^T \mathbf{X}$ is univariate normal for all $\mathbf{a} \in \mathbb{R}^d$.

PROP 5.7 If $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$ and Σ is positive definite, then $(\mathbf{X} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi_d^2$. Based on this fact, given some data $\mathbf{X}_1, \dots, \mathbf{X}_n$, we can assess the normality of the parent distribution by analyzing

$$(\mathbf{X}_i - \bar{\mathbf{X}})^T S^{-1} (\mathbf{X} - \bar{\mathbf{X}}) \stackrel{?}{\sim} \chi_d^2$$

Using the sample covariance matrix and sample mean, we can fit data to a multivariate normal. However, financial returns very rarely fit well.

PROP 5.8 If $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$ and $\mathbf{Y} \sim \mathcal{N}_d(\boldsymbol{\mu}^*, \Sigma^*)$ are independent, then

$$\mathbf{X} + \mathbf{Y} \sim \mathcal{N}_d(\boldsymbol{\mu} + \boldsymbol{\mu}^*, \Sigma + \Sigma^*)$$

A similar result holds when $\mathbf{X} \perp \mathbf{Y}$. (They are normal, but in not so nice of a form).

Pros and Cons

While normality is rarely a good model for financial returns (light tail, too much symmetry), there are a few advantages:

- Ubiquity
- Simulation is easy via $\mathbf{X} = \boldsymbol{\mu} + \Sigma^{\frac{1}{2}} \mathbf{Z}$.
- The characteristic function, density, and moments are all easy to work with.
- Linear combinations $\mathbf{a} + B\mathbf{X}$ remain normal (and so VaR is a coherent risk measure).
- Marginal and conditional distributions remain normal.
- Dependence structure is described entirely by Σ . Recall, for normally distributed variables, $\text{Cov}(X_i, X_j) = 0$ implies independence (this does not hold generally).

NORMAL VARIANCE MIXTURES

DEF 5.4 Let $\boldsymbol{\mu} \in \mathbb{R}^d$ and $A \in M_{d \times k}(\mathbb{R})$. In this context, we call these the *location vector* and *dispersion*

DEF 5.5 *matrix*. Let $\mathbf{Z}$ be a vector of k IID standard normals. Let W be a univariate, non-negative random variable, which is drawn independently of $\mathbf{Z}$. Then

$$\mathbf{X} \stackrel{d}{=} \boldsymbol{\mu} + \sqrt{W} A \mathbf{Z}$$

DEF 5.6 is called a *multivariate normal variance mixture*.

Eg. 5.1 Let $W \sim \text{Ig}(\frac{\nu}{2}, \frac{\nu}{2})$, the inverse gamma distribution. In other words, $\frac{1}{W} \sim \text{Gamma}(\frac{\nu}{2}, \frac{\nu}{2})$. Then $\mathbf{X} = \boldsymbol{\mu} + \sqrt{W} A \mathbf{Z} \sim t_d(\nu, \boldsymbol{\mu}, \Sigma)$ is a multivariate student-t.

The *symmetric generalized hyperbolic distribution* arises when $W \sim N^-(\lambda, \xi, \psi)$ is a generalized inverse Gaussian. We do not concern ourselves with the details of N^- .

DEF 5.7

If $\mathbf{X}$ is a normal variance mixture with $\mathbb{E}[W] < \infty$, then

PROP 5.9

$$\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu} \quad \text{cov}(\mathbf{X}) = \mathbb{E}[W]\boldsymbol{\Sigma} \quad \text{cor}(\mathbf{X}) = \mathcal{R}(\boldsymbol{\Sigma})$$

$\mathbb{E}[\mathbf{X}] = \mathbb{E}[\boldsymbol{\mu} + \sqrt{W}\mathbf{A}\mathbf{Z}] = \boldsymbol{\mu} + \mathbb{E}[\sqrt{W}] \cdot 0 = \boldsymbol{\mu}$, since $\mathbb{E}[\sqrt{W}] < \infty$. Similarly, we get

PROOF.

$$\text{cov}(\mathbf{X}) = \text{cov}(\boldsymbol{\mu} + \sqrt{W}\mathbf{A}\mathbf{Z}) = \text{cov}(\sqrt{W}\mathbf{A}\mathbf{Z}) = \mathbb{E}[\sqrt{W}\mathbf{A}\mathbf{Z}\mathbf{Z}^T\mathbf{A}^T\sqrt{W}]$$

Then, since $\sqrt{W}$ is univariate and independent, this is

$$\text{cov}(\mathbf{X}) = \mathbb{E}[W]\mathbf{A}\mathbb{E}[\mathbf{Z}\mathbf{Z}^T]\mathbf{A}^T = \mathbb{E}[W]\mathbf{A}\mathbf{I}\mathbf{A}^T = \mathbb{E}[W]\boldsymbol{\Sigma}$$

The correlation matrix comes out nicely, since

$$\text{cor}(\mathbf{X})_{i,j} = \frac{\text{cov}(X_i, X_j)}{\sqrt{\text{Var}(X_i)\text{Var}(X_j)}} = \frac{\mathbb{E}[W]\Sigma_{ij}}{\sqrt{\mathbb{E}[W]\Sigma_{ii}\mathbb{E}[W]\Sigma_{jj}}} = \frac{\Sigma_{ij}}{\sqrt{\Sigma_{ii}\Sigma_{jj}}}$$

□

Let $\mathbf{X} = \langle X_1, X_2 \rangle$ be a normal variance mixture with $\text{cov}(\mathbf{X}) = \mathbf{I}$, i.e. $\text{cov}(X_1, X_2) = 0$. Then $X_1 \perp X_2$ if and only if W is almost surely a constant. In other words, $\mathbf{X}$ must be normal.

PROP 5.10

Elliptical Distributions

We say that $\mathbf{X} \sim \varepsilon_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \psi)$ is an *elliptical distribution* if its character function satisfies

DEF 5.8

$$\phi_{\mathbf{X}}(\mathbf{t}) = \exp(i\mathbf{t}^T \boldsymbol{\mu})\psi(\mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t})$$

Eg. 5.2 When $\psi(x) = e^{-\frac{x^2}{2}}$, $\varepsilon_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \psi) = \mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. We conclude that the multivariate normal is an elliptical distribution.

Eg. 5.3 When $\psi(x) = \mathcal{L}_W(\frac{x}{2})$ is the Laplace transform with respect to the measure $dH(t)$, where $F_W = H$. We conclude that any normal mixture (multivariate student-t, generalized hyperbolic, etc) are elliptical.

When $\mathbf{X} \sim \varepsilon_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \psi)$, we have

PROP 5.11

1. If $B \in M_{k \times d}(\mathbb{R})$ and $\mathbf{b} \in \mathbb{R}^k$,

$$B\mathbf{X} + \mathbf{b} \sim \mathcal{E}_k(B\boldsymbol{\mu} + \mathbf{b}, B\boldsymbol{\Sigma}B^T, \psi).$$

2. If $\mathbf{X} = \langle X_1, X_2 \rangle$, with

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix},$$

along the obvious lines, then $X_1 \sim \mathcal{E}_k(\mu_1, \Sigma_{11}, \psi)$ and $X_2 \sim \mathcal{E}_{d-k}(\mu_2, \Sigma_{22}, \psi)$.

3. If $\mathbf{Y} \sim \mathcal{E}_d(\boldsymbol{\mu}^*, \Sigma, \psi^*)$ and $\mathbf{X} \perp \mathbf{Y}$, then

$$\mathbf{X} + \mathbf{Y} \sim \mathcal{E}_d(\boldsymbol{\mu} + \boldsymbol{\mu}^*, \Sigma, \psi\psi^*).$$

Theorem 5.1 Coherence of VaR over Elliptical Distributions

Let $\mathbf{X} \sim \mathcal{E}_d(\boldsymbol{\mu}, \Sigma, \psi)$. Let $\alpha \in [\frac{1}{2}, 1)$. Then, writing $\mathbf{X} = \langle X_1, \dots, X_d \rangle$, we have

$$\text{VaR}_\alpha(X_1 + \dots + X_d) \leq \text{VaR}_\alpha(X_1) + \dots + \text{VaR}_\alpha(X_d)$$

Thus, VaR_α is coherent.

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